Automated Classification, Detection, and Segmentation of Guava Diseases using Deep Learning

This project is submitted to the Department of Computer Science and Engineering, Dhaka International University, in partial fulfillment to the requirements of Bachelor of Science (B. Sc.) in Computer Science and Engineering (CSE).

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Declaration

We hereby declare that; this project "**Automated Classification, Detection, and Segmentation of Guava Diseases using Deep Learning**" submitted by Muhaimenul Islam, Roll No:- 02; Md. Rasel Mia, Roll No:- 18;Tabia sultana Nishi, Roll No:- 19;Md Kawser Sheikh, Roll No:- 25; Monira Akter, Roll No:- 28; has been carried out by us and it has been submitted for the award of the B.Sc. degree. We also certify that this project was prepared by us for the purpose of fulfillment of the requirements for the Bachelor of Science (B.Sc.) in Computer Science and Engineering.

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Dedicated to

Our Parents
&
Teachers

Abstract

Guava fruits and leaves are beneficial to the economy, but they can also easily make people ill, which can reduce production and even affect public health. Since the appearance of a disease symptom can be different in different regions, it can be difficult to determine what illness you have. This research aims to suggest a complete deep learning system for the identification, classification, and segmentation of diseases in guava leaves and fruits. This will enable us to determine what illnesses people have quickly and correctly. We split the dataset into five portions and employed preprocessing, protocol-based approaches, and GAN approaches to increase the number of images in the dataset by 7,000. This increased the stability and applicability of the model. We considered a number of deep learning models that could be employed for disease classification, such as DenseNet121, NASNetMobile, MobileNetV2, InceptionV3, Xception, ResNet50DenseNet121_Fusion, EfficientNetDualBlock CNN. Among these models, the fusion model of ResNet50 and DenseNet121 had the highest accuracy of 98%. We employed YOLOv8-seg and Mask R-CNN for object detection and separation. The best-performing model for lesion detection was YOLOv8-seg. The mAP@0.5 was 0.907, mAP@0.5:0.95 was 0.783, precision was 0.899, recall was 0.88, and F1-score was 0.879. We used Explainable AI (XAI) methods on the best models to make them more practical in the real world. There was also a web app where individuals could search, sort, and categorize diseases as they occurred. The system is easy to use, explain, and work with, especially for devices with limited resources. Finding diseases quickly is good for the economy and public health.

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Chapter 1 Introduction

1.1 Overview

People often refer to guava (*Psidium guajava*) as the "poor man's apple" or the "apple of the tropics." The leaves and fruit of guava are both very healthy because they are high in potassium, fiber, antioxidants, and vitamin C. Guava is very good for your health because it has nutrients that help your immune system, protect you from infections, improve digestion, keep your heart healthy, and lower high blood pressure [2]. Guava does well in both tropical and subtropical places [3]. Guava fruits and leaves are prone to various diseases that diminish yield, causing economic losses for farmers and health hazards for the populace.

1.2 Problem Definition

Diseases that affect guava fruit and leaves have a big impact on the country's economy. Bangladesh grows guava on almost 0.3 million hectares and makes about 3.7 million metric tons of it every year. This makes the country one of the top guava producers [4]. According to FAO data, Bangladesh is the eighth largest producer of guavas in the world and the tenth largest producer of all fruits [4]. The main things that hurt guava fruits and leaves are fungi, bacteria, and viruses [3]. Guava diseases appear in various forms due to differences in cultivars and weather.

1.3 Factor of The Problems

Farmers used to try to figure out what diseases were by looking at them, which was often hard to do correctly. It was hard to figure out what was wrong with guava plants because of their shape, symptoms, leaf color, variety, and differences in the environment [6]. Right now, spectrometers and real-time polymerase chain reaction (PCR) are used to find diseases. However, these methods are slow, need skilled workers, cost a lot of money, and are hard to understand. These problems can be solved with machine learning, deep learning, and computer vision. They help you find diseases, figure out where they are, and figure out how big the area that is sick is [5]

1.4 Traditional Approach

Many studies have employed machine learning and deep learning techniques to analyze guava diseases. The Inception–ResNet50 model was able to classify guava leaf disease with an accuracy of 97.50% [7]. A multi-fusion meta-learning model attained 97.32% accuracy in identifying and categorizing guava leaf disease, exceeding ensemble machine learning techniques [8]. A deep learning framework that used EfficientNetV2 and Vision Transformer got 95% accuracy. It also had a web app and GradCAM-based XAI [9]. Another study used EfficientNetB3 and got 94.93% accuracy in predicting and classifying diseases [5]. Models used for segmentation-based detection were GLSM (83.40%), YOLOv5 (73.3%), and GIP-

MU-Net (92.4%) [6]. DenseNet201 the highest accurate of 96%, it came to classifying diseases in Guava fruit [10].

1.5 Contribution summary

This research introduced a deep learning framing for the identification, detection, and segmentation of diseases in guava fruit and leaf. We used both popular and GAN-based methods to preprocess and improve the dataset for classification. This dataset problem an unbalanced, we used completed sampling, strati form sampling, and fake image generation. We check it and a lot of deep learning models, and DenseNet121 the best models. We used explainable AI (XAI) means to make easier to understand the models. We used Roboflow to prepare triangle-based annotations for detection and segmentation, and we used Albumentations to do augmentation that was aware of segmentation. We tested several models, with YOLOv8-Seg and Mask R-CNN. Pixel-level masks were made from YOLO triangle annotations for training semantic segmentation. The best performance of the YOLOv8-Seg, find the objects and separate pixels at the time. Finally, the system was set up as a web app that keep find, identify, and divided diseases from beginning to end.

1.6 Objectives

- Image preprocessing, popular, and GAN-based image augmentation means used to improve data range and power. Class imbalance problems addressed using stratified sampling, oversampling, and image synthesis.
- Single model for segmentation-based guava fruit and leaf disease diagnosis, eliminating the need for separate models. Digitally labeled polygons and segmentation-driven image augmentation helped in accurate localization.
- Polygon-annotated YOLO datasets in COCO JSON format for training the Mask R-CNN model. Instance segmentation masks are created using the annotations in the dataset, consistent with the task.
- Contour-based post-processing method with lower computational complexity for edge devices. Pixel-level infection area estimation is useful for determining the severity of the disease.
- Grad-CAM was used to enhance the transparency, interpretability, and trustworthiness of the model. Developed a user-friendly web application for the recognition, detection, and segmentation of diseases from a single image.

The structure of this paper is as follows: Section 1 discusses why guava fruit and leaf diseases are a problem. Section 2 describes guava diseases, their causes, and types. Section 3 shows the statistical analysis of the problem. Section 3 presents the statistical analysis of the problem. Section 4 reviews related work and existing models. Section 5 discusses the proposed methods, models used, and their performance. Finally, Section 6 presents the contributions of this research.

Chapter 2

Literature Review

2.1 Basic Analysis

Recently, significant work has been done on guava fruit and leaf recognition, detection, and segmentation using deep learning and machine learning models. Deep learning architectures such as InceptionV3, ResNet50, ViT-B16, EfficientNetV2, GoogLeNet, VGG19, DenseNet201/169, MobileNet, Xception, NASNetLarge, U-Net + SegNet, VGG16, and Mask R-CNN have shown high accuracy and strong performance, proving their effectiveness for automated guava disease analysis.

2.2 Related Works

Osman Guler et al. [7] proposed an ensemble deep learning framework for guava leaf disease classification using hybrid data augmentation. The study utilized 2,063 images, divided into five categories, and applied seven deep learning models, including InceptionV3, ResNet50, and ViT-B16. InceptionV3 achieved 92.50% accuracy on GAN-generated data, while ResNet50 obtained 93.12% accuracy on augmented data. The ensemble of InceptionV3 and ResNet50 produced the best results with 97.50% test accuracy, 0.975 F1 score, and 0.9934 AUC . M. Asim et al. [8] guava leaf disease detection and classification, a computational intelligence–based framework was proposed using ensemble machine learning and a multifusion metalearning model. The study employed KNN, SVM, ANN, and Random Forest classifiers. A total of 560 image samples were used, and data preprocessing techniques such as augmentation, resizing, scaling, and normalization were applied. The framework detected seven types of guava leaf diseases. Experimental results showed that the multi-fusion meta-learning model outperformed the ensemble machine learning model, achieving an accuracy of 97.32%. Sarah Farooqui et al. [9] proposed a deep learning–based method for guava leaf disease detection and classification using five disease classes. The dataset was split into 80% training and 20% testing, and evaluation was performed on original images. A hybrid model combining EfficientNetV2 and Vision Transformer (ViT) achieved 95% accuracy. Using Grad-Cam to highlight infected places of leaves describing the explainable study. Also for enabling disease detection, explanation and solution access a web based deployment system they integrated. Oya at al. [11] used the machine learning process and deep learning, it detects guava fruit disease. They used totally 473 images to classify anthracnose, fruit fly infected and healthy categories. At preprocessing, image segmentation, CLAUHE and data augmentation were applied. In those models, SVM and ANN obtained the best performance with 99% accuracy, which was confirmed, while using confusion matrix that showing minimal misclassification. Asad S. Dowtom at al. [5] used deep learning model for guava disease classification and their prediction. There are total 1834 images in dataset and they are divided into training (1239), validation (457), and testing (138) sets. Also here rotation, translation, cropping and noise addition were applied for image preprocessing and data augmentation. Evaluated total 4 pre-trained CNN models and EfficientNetB3 obtained the best overall performance with a test accuracy of 94.93%. Javid Rashid et al. [6] proposed a hybrid deep learning framework to detect multiple diseases on guava leaves in three stages: GIP-MU-Net

for infected patch segmentation, GLSM for leaf segmentation, and GMLDD (YOLOv5) for detecting multiple diseases on a single leaf. The study used two datasets— Guava Patches and Guava Leaf Diseases—comprising 2,542 images with 6,241 annotations across five disease classes. Preprocessing of data augmentation applied to enhance model performance. Results showed GIP MU-Net achieved 92.41% accuracy, GLSM 83.40% accuracy, and GMLDD obtained 73.3% precision, 73.1% recall, 71.0% mAP@0.5, and 50.3% mAP@0.5:0.95. Buket Toptas et al. [10] A pre-trained CNN model was used for guava fruit classification. The dataset contains 2,310 images of three guava varieties. and three ripeness levels, which are divided into nine classes. Both k-fold stratification and 80:20 split were used. Among Google Net, VGG19, ResNet50, and DenseNet201, DenseNet201 performed best with accuracy 0.9650, sensitivity 0.9677, specificity 0.9954, F1-score 0.9681, MCC 0.9640, and Kappa 0.8268.

Deepak Thakur et al. [8] proposed a hybrid deep learning framework for the prediction and classification of healthy and diseased coffee leaves. They used JMUBEN and JMUBEN2 datasets, retaining 57,275 images after cleaning. Data augmentation increased the dataset to 572,750 images. The SUNet model combines UNet + SegNet, VGG16, and Mask R-CNN, achieving performance improvements of 1.22% accuracy, 1.21% IoU, 1.17% F1-score, 1.19% precision, and 1.24% recall. Nidhi Parashar et al. [13] proposed a CNN-based Inception-v3 model for apple leaf disease detection. To evaluate the model's performance, 5-fold cross-validation was applied. A total of 1,821 apple leaf images were used, divided into four distinct classes with no class overlap. Image preprocessing and data augmentation techniques were employed to enhance model performance. The proposed model achieved an accuracy of 94.76%, precision of 84.60%, recall of 87.40%, and an F1-score of 85.00%. Ibrahim Çetiner et al. [14] used a CNN-based deep learning model for apple leaf disease classification using computer vision preprocessing techniques and deep learning methods. A dataset of 3,651 images was divided into four classes and split into 80% training and 20% testing after cleaning. Pre-trained models such as VGG19, DenseNet169, MobileNet, Xception, and NASNetLarge were used as input layers, and the proposed CNN achieved 98% accuracy. Roshni Polly et al. [15] Deep learning models have been used for classification, detection, and segmentation of leaf diseases. Accurate and reliable disease identification was achieved by integrating CNN-based classifiers with advanced models such as YOLOv8 for detection and DeepLabV3+ and U-Net for segmentation. The proposed method demonstrated robust performance, achieving an overall accuracy of 97%, confirming its effectiveness in analyzing specific leaf diseases. Muhammad Umar et al. [16] The deep learning models were applied for classification and detection of tomato leaf diseases. They worked on seven different types of tomato leaf diseases, applying CNN for classification and YOLOv7 for detection of the diseases. The CNN model was able to strongly classify the diseases with the highest achieve of 98.8%, identifying its strength and consistency in correctly detecting the tomato leaf diseases. Md. Manowarul Islam et al. [17] developed a plant leaf disease detection system using a large dataset consisting of 10,000 leaf images. Several deep learning models including VGG-16, VGG-19, CNN, and ResNet50 were used for disease classification. Among these models, ResNet50 achieved the

best performance with 98.60% accuracy. Furthermore, the trained ResNet50 model was integrated with a web-based application, which enabled effective differentiation between healthy and diseased leaves, as well as accurate classification of various plant leaf diseases. Shah Faisal et al. [18] developed an automated deep learning-based system for rapid classification and detection of citrus tree diseases. Their dataset initially consisted of 759 images, which were expanded to 3,383 images using image preprocessing and data augmentation.This studyapplied transfer learning with pre-trained CNN models such as EfficientNetB3, ResNet50, MobileNetV2, and InceptionV3, where EfficientNetB3 achieved the highest performance with training, testing, and validation accuracies of 99.43%, 99.48%, and 99.58%, respectively. Yohannes Agegnehu Bezabh et al. [19] developed an ensemble CNNbased model for mango leaf disease detection.An initial dataset of 1,452 images was expanded to 3,195 images using preprocessing and augmentation techniques. Based on GoogLeNet–VGG ensemble model achieved 99.87% training accuracy, 99.72% validation accuracy, and 99.21% testing accuracy. Adnan Haider et al. [20] Segmentation architectures are used to accurately and rapidly detect leaf and fruit diseases. In this study, EFFS-Net and MDFS-Net models were applied to three datasets: BRACOL coffee leaf dataset (500 images), apple orchard fruit dataset, and cassava root disease dataset (1036 images). The proposed models achieved Dice similarity coefficients of 88.81%, 95.01%, and 86.04%, demonstrating effective and computationally efficient disease segmentation performance. Abebech Jenber Belay et al. [21] A deep learning-based model was used for chickpea disease detection. Image preprocessing and data augmentation techniques were applied, the dataset was increased from 1,399 images to 8,391 images. In this study, a combined CNN and LSTM model was used for disease detection, which achieved a high accuracy of 92.55%. Chunduri Madhurya et al. [22] developed an integrated and optimized deep learning framework for fast and accurate plant leaf disease detection and classification. In their study, image preprocessing and data augmentation techniques were applied to improve the performance of the model. The proposed framework, named YR2S, combines YOLOv7, PCFAN, ShuffleNetV2 and a hybrid optimization approach, achieving a high accuracy of 99.69% in disease detection and classification. Nagamani H. S. et al. [23] Fuzzy-SVM, CNN, and R-CNN models were used for tomato leaf disease detection. The models were applied to a dataset of 735 leaf images to determine the most accurate method. Among these models, the R-CNN-based classifier achieved the highest accuracy of 96.935%, outperforming other methods. Kazi Naimur Rahman et al. [24] Deep learning was applied to identify plant leaf diseases using a dataset of 30,945 images from eight plant types and 35 disease classes. A CNN model achieved 95.62% accuracy in leaf classification. Nine CNNbased models (custom CNN, VGG16, VGG19, InceptionV3, MobileNet, DenseNet121, Xception, and two hybrid models) were tested, achieving up to 100% accuracy for various crops such as potato, bell pepper, apple, and peach, demonstrating highly effective disease detection performance. Muhammad Ahmed et al. [25] Using Deep Learning (DL) and Transfer Learning (TL), ResNet-101 model was used to detect Guava disease. They prepared a dataset of 3,784 real images, which were then processed

through image preprocessing and augmentation to produce 4,632 images. The ResNet-101 model achieved an impressive accuracy of 98.48%, and Grad-CAM was used to ensure that the network focused on the infected areas. Kuo-Dung Chiou et al. [26] used the YOLOv4 pertained network architecture to detect external defects and damages in guava fruits. They collected 189 guava fruits and, through image preprocessing and augmentation, created 1,701 images. The YOLOv4 model achieved a false positive rate of 6.62%, a false negative rate of 5.03%, and an accuracy of 88.15%. Reqia Mirjat et al. [27] used image processing and machine learning techniques to detect wilt disease in guava leaves at its early stages. They created a dataset of 1,420 guava leaf images and divided it into three categories based on the level of noise. The segmentation of guava wilt disease was performed using Kmeans clustering and CNN techniques, achieving accuracies of 93.82% and 95.17%, respectively. Prabhjot Laur et al. [28] used machine learning algorithms for disease classification in grape plants. Their research focused on two out of the four identified diseases. In this study, the EfficientNet B7 architecture was employed with transfer learning, achieving an accuracy of 98.7% after 92 epochs. Fizzah Arshad et al. [29] developed a hybrid deep learning model named PLDPNet for detecting potato leaf diseases. They divided 2,152 images into three classes. In this study, they used an ensemble approach combining VGG19 and Inception-V3 models and employed Vision Transformers in the hybrid framework. By utilizing segmentation and fusion features, the PLDPNet model achieved an accuracy of 98.66% and an F1-score of 96.33%. Anwar Abdullah Alatawi et al. [30] discussed the importance of machine learning for plant disease detection and proposed a deep learning model. They selected 15,915 leaf images covering 19 different types of plant diseases and applied the proposed VGG16 model, achieving an accuracy of 95.2%. Hassan Ali et al. [31] fine-tuned the EfficientNet-B0 model, achieving 99.69% accuracy on the APV dataset and 99.78% on the PV dataset, using data augmentation, class weighting, and an IoU threshold of 0.5 to address class imbalance. Sunil S. Harakannanavar et al. [32] a model using K-means clustering, DWT, PCA, and GLCM for feature extraction, and CNN for classification. This model achieved 99.09% accuracy in detecting tomato leaf diseases, outperforming existing methods with high precision and recall in six leaf diseases. M. A. R. Nishad et al. [33] The focus is on potato leaf disease detection and classification using Kmeans clustering and deep learning algorithms. The VGG16 model achieved the best results, reaching 97% accuracy after applying data augmentation techniques and K-means partitioning. N.Mamat et al. [34] YOLOv5 was used for automatic fruit classification, achieving 99.5% accuracy on a dataset of 3,500 fruit images with a threshold of mAP@0.5. The method demonstrated high efficiency and accuracy in the annotation of oil palm fruits and other fruits through iterative annotation tasks. Yuan et al. [35] proposed an optimized DeepLab V3+ model for the segmentation of black rot spots on grape leaves. The proposed model was tested on 1,180 images, including indoor and garden field images, and achieved better segmentation results with mIoU = 0.848 and F1-score = 0.918, outperforming the original DeepLab V3+ model. Faiza Khan et al. [36] Developed a mobile-based system for detecting corn diseases, using deep learning models such as YOLOv3-tiny, YOLOv4, YOLOv5s, YOLOv7s, and

YOLOv8n. The YOLOv8n model achieved the highest accuracy of 99.04% for detecting and classifying corn diseases, including blight, leaf spot, and sugarcane mosaic virus. Shafik et al. [37] The ResNet-9 model was used for plant disease classification, achieving 97.41% accuracy on the TPPD dataset, which contains 4,447 images in 15 classes. The model used data augmentation and hyperparameter optimization to handle class imbalance and demonstrated improved interpretability with SHAP saliency maps, demonstrating IoU > 85% for feature alignment. Andrew J. et al. [38] Deep learning models were used to classify 38 plant diseases from the PlantVillage dataset, which contains 54,305 images. The study used transfer learning with VGG-16, ResNet-50, Inception V4, and DenseNet-121, achieving 99.81% accuracy with DenseNet-121.

Mukhiddinov et al. [39] Developed an improved YOLOv4 model to classify fruits and vegetables as fresh or rotten, achieving 75.8% training accuracy and 73.5% experimental accuracy using a dataset of 12,000 photos. The model has been released over previous YOLO versions and has demonstrated performance with data augmentation technology for increased classification accuracy. Thanh-Hai Nguyen et al. [40] A VGG-19 model with transfer learning and HSV image segmentation is proposed to classify tomato leaf diseases, achieving 99.72% accuracy. The model trained on 16,010 images from the PlantVillage dataset has improved its efficiency classification and minimized training time by concentrating on the leaf area and employing a black background for all images. Mimma et al. [41] YOLOv3 and YOLOv7 were used for fruit detection and ResNet50 and VGG16 for classification. The proposed model achieved 99% accuracy using ResNet50 and 98% accuracy using VGG16 on a custom dataset of eight classes of fruits. The results show high accuracy for both fruit detection and classification. Zayani et al. [42] Tomato a YOLOv8-based deep learning model for disease detection is proposed, which achieved 66.67% accuracy after training on an extended RoboFlow dataset. The performance of the model varies according to disease class and future work is suggested to address data imbalances and explore alternative architectures. Nouman Butt et al. [43] A deep learning approach combined with Moth-Flame has been introduced to optimize citrus disease detection, achieving 99.6% accuracy using transfer learning with models such as DenseNet-201 and AlexNet. The proposed hybrid framework outperforms existing methods and addresses the challenges of feature selection and classification in citrus cultivation. Christian A. Elinisa et al. [44] A U-Net model is proposed for early detection of Fusarium wilt and Black Sigatoka in banana, which achieved 96.45% Dice coefficient and 93.23% IoU for image segmentation. The model, trained on 18,240 images, accurately segmented the affected areas of banana leaves and stems. Rijon et al. [45] It was focused on the detection of guava diseases using a CNN-Gradient Boosting ensemble model with 99.99% accuracy. The proposed method uses EfficientNet-B0 as the feature extractor and machine learning algorithms such as Gradient Boosting for disease classification, including Fruit Flies and Anthracnose.

Table 2.1: Summary on Existing Work on Fruit and Leaf Disease

Author	Class	Methods	Accuracy (%)
Assad Doutoum et al. [5]	-	Four pretrained CNN models.	94.93%
Javed Rashid et al. [6]	5	GIP-MU-Net, GISM, GMLDD	96.76%
Osman Guler et al. [7]	5	Seven Deep Learning Models, including inception, restNet50, VIT-B16	97.50%
Muhammad Asim et al. [8]	7	KNN, SVM, Random Forest, Multi-Fusion Meta Learning Models	97.32%
Sarah Farooque et al. [9]	5	EffeciencyNetV2 + Vision Transformer, Grad Cam Web application	95%
Buket Topas et al. [10]	9	GoogleNet, VGGL9, RestNet50, DenseNet20	92.41%
Nidhi-Parashar et al. [13]	4	CNN Based Inception V3	94.76%
Ibrahim Centiner et al. [14]	4	VGGL9, DenseNet69, Mobile Net, Xception, NasaNet Large.	98%
Proposed Method	5	DenseNet121, NASNetMobile, MobileNetV2, InceptionV3, Xception, ResNet50, DenseNet121_Fusion, and EfficientNetDualBlock_CNN, YOLOV8-seg,MASK R-CNN	98% and mAP@0.5 of 0.907, mAP@0.5:0.95 of 0.783, precision of 0.899, recall of 0.88, and F1-score of 0.879

Chapter 3

Methodology

3.1 Overview

The proposed project work focuses on the development of a combined deep learning solution for the analysis of guava leaf and fruit diseases, which consists of three tasks: classification, detection, and segmentation. The aforementioned workflow includes data acquisition, preprocessing, splitting of the dataset, and dataset augmentation, along with training and analysis. For classification, there were used Hybrid Models and Pertained CNN. Using annotated bounding boxes, the object detection models are localized infected region, while the sematic segmentation provides pixel-level disease identification. Using Accuracy, mAP, IoU, and Dice coefficient, the performance was evaluated. Lightweight and deep models were compared for their deployment, feasibility and accuracy. Finally, XAI was applied to the model to ensure that it is explainable and that decisions are consistent.

3.2 Conceptual Architecture of the Proposed Approach

The proposed architecture presents a comprehensive deep learning architecture that aims to automatically analyze the diseases that affect the guava leaves and fruits. The proposed system presents a step-by-step approach that begins with data preparation. In this phase, to ensure stable model training, first of all, raw images are aggregated and then preprocessed, augmented, and balanced. The enhanced dataset is beneficial in that it enables the model to learn robust features of the model with reduced issues caused by class imbalance.

For classification, convolutional neural networks are employed to classify the diseases. A hybrid fusion model that combines ResNet50 and DenseNet121 is employed to enable the system to learn overall structural information as well as detailed lesion information, thus enhancing the accuracy of the overall disease prediction. After classification, the detection and segmentation module recognize diseased zones while creating pixel-wise masks. YOLOv8-Seg is generally used for real time detection and mask generation, that improves overall quality and accuracy of segmentation while the Mask R-CNN helps to refine object boundaries. To enhance transparency, Explainable AI techniques such as Grad-CAM, LIME, SHAP, SmoothGrad, and saliency maps are integrated, allowing visualization of decision-relevant regions.

Finally, the optimized models are deployed using a web-based platform, which supports real-time classification, detection, segmentation, and visualization of the models across various devices for agricultural applications.

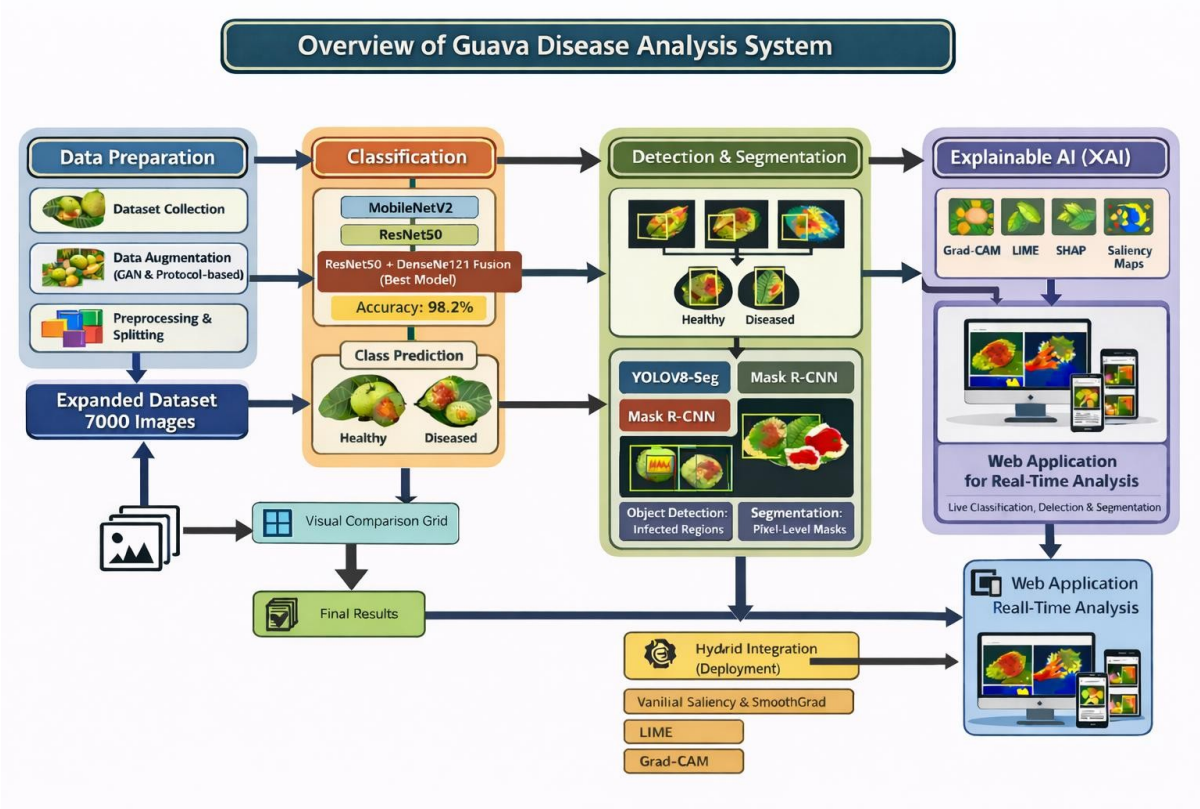


Figure 3.1: Overall system architecture for guava fruit and leaf disease

3.3 Dataset Description

3.3.1 Dataset for Classification

The dataset used in this study is obtained from Kaggle [46] and includes images of guava leaf and fruit diseases. The dataset includes a total of 527 images, which are classified into five categories: Disease Free (126 images), Phytophthora (114 images), Red Rust (87 images), Styler and Root (94 images), and Scab (106 images). The dataset was initially imbalanced, meaning that the number of images in each category was not equal, as shown in Figure 3.2.

The dataset was split into Training, Validation, and Testing datasets using a stratified split of 70%-10%-20%. The splitting was done while allowing the data to be shuffled to remove any sequential bias. After splitting, the Training dataset had 369 images, the Validation dataset had about 50 images, and the Testing dataset had about 111 images. The Validation and Testing datasets were left untouched to avoid any bias in the evaluation of the model performance.

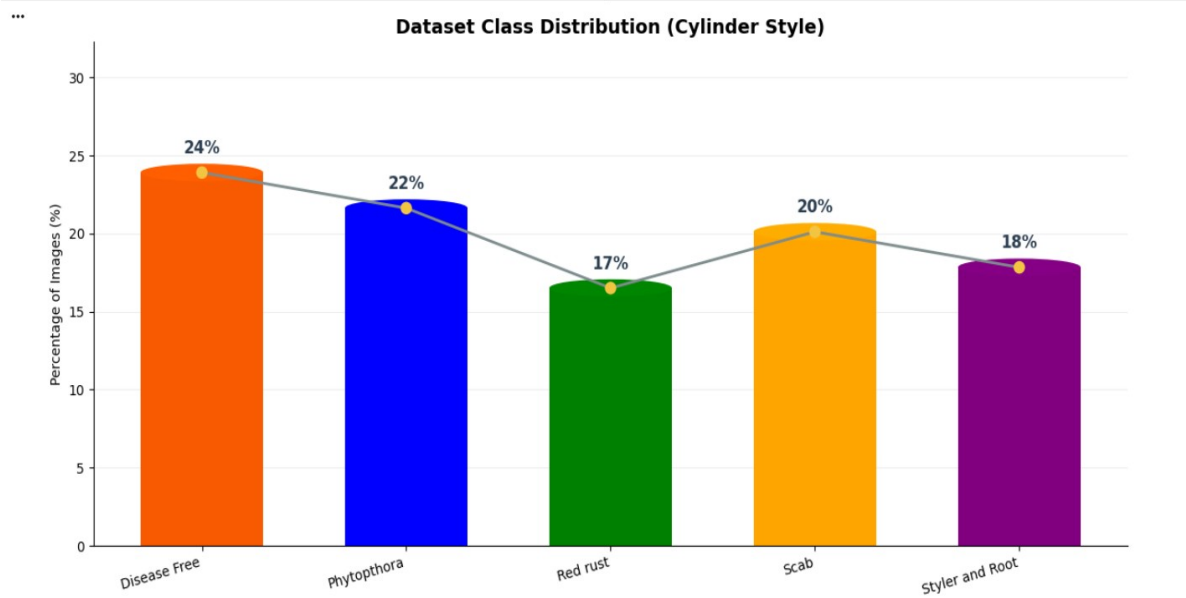


Figure 3.2: Raw Dataset Class Distribution

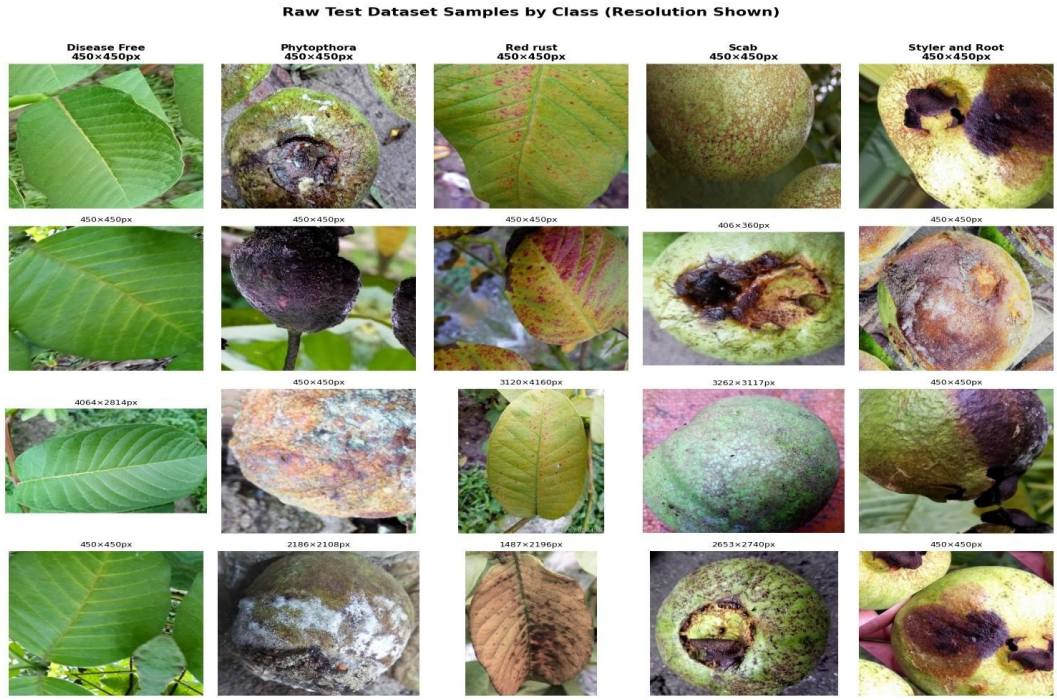


Figure 3.3: Raw Dataset Sample Images per Class

Image preprocessing was performed using ImageDataGenerator and flow_from_directory(). All images were resized to 224×224 pixels and normalized by scaling pixel values to the range of 0–1 (rescale = 1/255). Class labels were converted into categorical (one-hot encoded) format for multi-class classification. These preprocessing steps were applied to the Training, Validation, and Test datasets. However, augmentation was applied only to the Training set.

Since the Training dataset was imbalanced, balancing was applied only to the Training set using augmentation and oversampling techniques. This balancing strategy ensures that the model learns representative features from all classes equally. Each class was expanded to 1400 images, resulting in a total of 7000 training images(1400 × 5 classes) shown in figure 3.4. Data augmentation techniques included random rotation, zooming, and horizontal flipping to improve generalization and reduce overfitting. Validation and Test datasets were not augmented, ensuring scientific validity and unbiased performance evaluation. Based on the balancing target, each class was expanded to 1400 images, making the final Training dataset both balanced and augmented. No augmentation or balancing was applied to the Validation and Test datasets. Only resizing and normalization were performed.

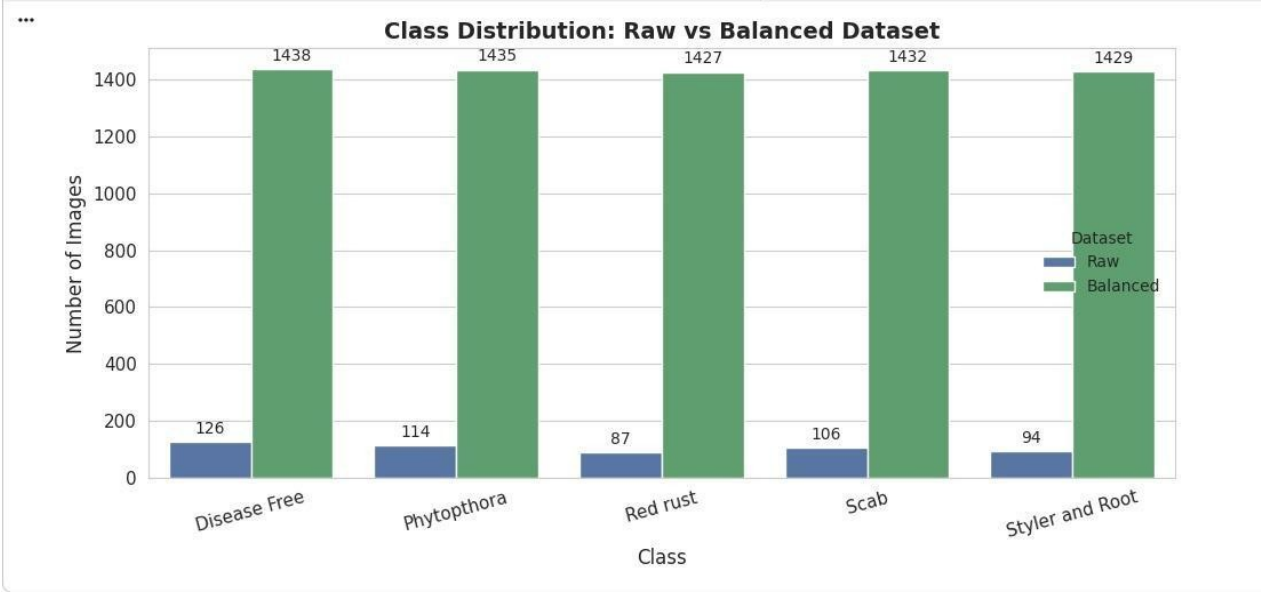


Figure 3.4: Preprocessing and Augmentation Comparison Dataset

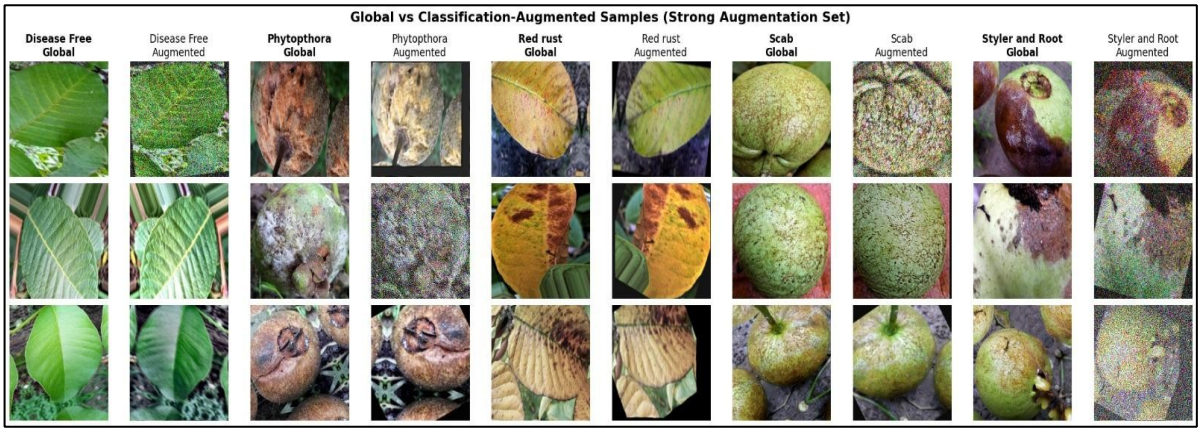


Figure 3.5: Original vs Augmented (Augmented samples)

Table 3.1: Summary Table of Dataset at Each Stage

Train Class	Original (70%)	Validation (10%)	Test (20%)	Balanced Train	Augmented Train
Disease Free	126	88	13	25	1400
Phytophthora	114	80	11	23	1400
Red Rust	87	61	9	17	1400
Styler and Root	94	66	9	19	1400
Scab	106	74	11	21	1400
Total	527	369	53	105	7000

3.3.2 Dataset for detection

We used the guava fruit and leaf disease dataset to find areas where the disease was affecting objects. A single experimental framework was kept in place so that the same dataset could be used for image-level classification, region-level detection, and pixel-level segmentation at all times. This helps to ensure that the comparison between different learning paradigms is done equally well while still considering the differences in lighting, background, and lesions in the real world. The test annotations were kept completely separate from the training procedure to make sure that there were no leaks in the evaluation.
Roboflow was employed for manually annotating all the images. While annotating images for object detection, bounding boxes were placed around only the visible healthy or diseased parts. To avoid any mistakes in the annotations, overlapping lesions or lesions that are not clearly visible were not taken into account. While performing the segmentation task, polygon annotations were employed to outline the lesions at the pixel level. This made it much easier to identify the lesions precisely, which helped both the segmentation models and the subsequent severity evaluation. The dataset was saved in YOLO format with normalized coordinates to ensure that it was compatible with modern object detection architectures. Before training, it was ensured that the annotations were consistent, the labels were correct, and the coordinates were normalized correctly. The images and their labels were also checked to ensure that the formatting and normalization were correct.

To fix the class imbalance, only the training set was added to and balanced to 1400 images per class (for a total of 7000 images). We used Albumentations with the Ultralytics YOLO framework to do the augmentation. When the changes included flipping, changing the brightness and contrast, then also changing the hue and saturation, and rotating and scaling the image For the segmentation task, the normalized coordinates were first converted into pixel values so they could be accurately placed onto the images. After that, they were converted back into normalized form to keep the labelling format consistent. It helped to

maintain accurate alignment between the annotations and the images, it is also ensuring the training dataset which are remained balanced.

3.4 Applied models

3.4.1 Models for Classification

This study evaluated various deep learning models for the classification of diseases in guava leaves and fruits, including DenseNet121, NASNetMobile, MobileNetV2, InceptionV3, Xception, a combined ResNet50–DenseNet121 model, and an EfficientDualBlock CNN. These models were chosen because of previous research on the detection of diseases in plant leaves and fruits that produced favorable results, often reaching high accuracy

3.4.1.1 DenseNet121

DenseNet (DenseNet121) was chosen because of its effective feature reuse and gradient flow, which makes it more capable of generalizing on small to medium-sized datasets. The model has 121 layers and takes in 224×224×3 inputs. The layers are grouped into four dense blocks. Transfer learning was used by freezing the pretrained backbone and adding a custom classification head with about 8 million parameters [47].
The model was trained using categorical cross-entropy loss**:**

$$L = -\sum y_i \log(\hat{y}_i) \quad \ldots.(1)$$

And improved with Adam (1e-4), EarlyStopping, and lowering the learning rate. DenseNet121 is a good size (about 30MB) and has a good inference speed for real-world use.

3.4.1.2 NASNetMobile

We chose NASNet (NASNetMobile) because it is a lightweight architecture made possible by Neural Architecture Search (NAS), which cuts down on FLOPs and makes parameters more efficient [51]. It takes in 224×224×3 input and is made up of Normal and Reduction cells that were found on their own. It has about 5.3 million parameters**.** The computational cost of depth wise separable convolution is:

$$Cost_DW = Dk^2 \cdot M \cdot Df^2 + M \cdot N \cdot Df^2 \quad \ldots.(2)$$

Transfer learning was applied with the backbone frozen and optimized using Adam. Due to its small size (~23MB), NASNetMobile is highly suitable for mobile and edge deployment, particularly for smartphone-based agricultural applications.

3.4.1.3 MobileNetV2

MobileNetV2 was selected as a strong lightweight baseline model [18]. It utilizes inverted residual blocks and linear bottlenecks with depth wise separable convolutions [51]. The architecture takes 224×224×3 input and contains approximately 3.4 million parameters.

The residual block formulation is:

$$y = x + PW(DW(PW(x))) \quad \ldots\ldots(3)$$

Transfer learning was applied with Adam (1e-4) and EarlyStopping. With a very small size (~14MB) and fast inference, MobileNetV2 is highly feasible for real-time edge deployment.

3.4.1.4 InceptionV3

InceptionV3 was chosen for its multi-scale feature extraction capability. The model processes 224×224×3 input using parallel convolutional filters (1×1, 3×3, 5×5) within Inception modules and contains approximately 23 million parameters

Parallel feature concatenation is defined as:

$$x = [\, f1 \times 1(x) , f3 \times 3(x) , f5 \times 5(x) \,] \quad \ldots\ldots..(4)$$

Although computationally heavy (~90MB), InceptionV3 acts as a powerful feature extractor and is more suitable for server-side deployment [50].

3.4.1.5 Xception

Xception (Extreme Inception) uses depthwise separable convolutions to learn features quickly and easily [14]. The architecture has about 22 million parameters and is made up of Entry, Middle, and Exit flows.

The separable convolution operation is expressed as:

$$Conv_sep = Conv_depthwise + Conv_pointwise \quad \ldots\ldots\ldots..(5)$$

Xception maintains high feature extraction capability while reducing computational cost. Since it has a larger architecture compared to many other models, it is more suitable for cloud-based inference rather than lightweight deployment.

3.4.1.6 EfficientDualBlock_CNN

The EfficientDualBlock_CNN is a light-weight, specially designed architecture that can classify diseases in real-time. It has an input size of 224×224×3 and consists of three dual blocks with filter sizes of 32→64→128. These dual blocks are comprised of standard convolution and depth-wise convolution branches, feature addition, and maximum pooling

[48]. The final layers of the network comprise The last layers of the network comprise global average pooling, dense (256), dropout, and softmax. It was designed from scratch and optimized using the Adam optimizer, unlike other models that use pre-trained weights. This model is particularly useful for mobile and edge computing applications, which give high priority to parameter pruning, feature aggregation, and rapid processing.

3.4.1.7 Combining ResNet50 and DenseNet121

The combination model integrates the residual learning capabilities of ResNet (ResNet50) with the dense feature reuse capabilities of DenseNet (DenseNet121). Two pre-trained backbones take the same 224×224×3 input simultaneously. After the Global Average Pooling layer, the features are aggregated and then connected to the Dense (512), Dropout (0.5), and Softmax layers. Residual learning is defined as:

$$Y = F(x) + x \quad \text{…….. (6)}$$

This combination approach provides feature vectors that complement each other well and provide the highest accuracy in classification. However, because of its higher computational cost, it is more suitable for server-based agricultural advisory systems [49].

3.5 Best Model Architecture:

The ResNet50–DenseNet121 fusion network is a two-backbone convolutional model that can find diseases in RGB images of fruits and leaves that are 224 × 224 × 3 pixels. It uses both architectures to classify the picture in the end. We use Global Average Pooling to turn the outputs of both backbones into short feature vectors. After this step, the vectors are set together and it makes a useful pictures. ResNet50 learns more about the meaning and structure things, while the DenseNet121 learns more about small spot patterns and well defined textures. We use Global Average Pooling to turn the feature maps from both networks into small vectors. Then, we combine them to make a full fused representation. This vector goes through a fully connected layer with dropout regularization and then a softmax layer, which gives the probabilities of each disease class. We use frozen backbones, the Adam optimizer, and categorical cross-entropy loss for training. Early stopping and learning rate scheduling are used to prevent overfitting of the model. This fusion model integrates residual learning and dense feature reusing to achieve more stable, accurate, and diverse feature extraction, which is better than single-backbone models in terms of overall high accuracy disease assessment.

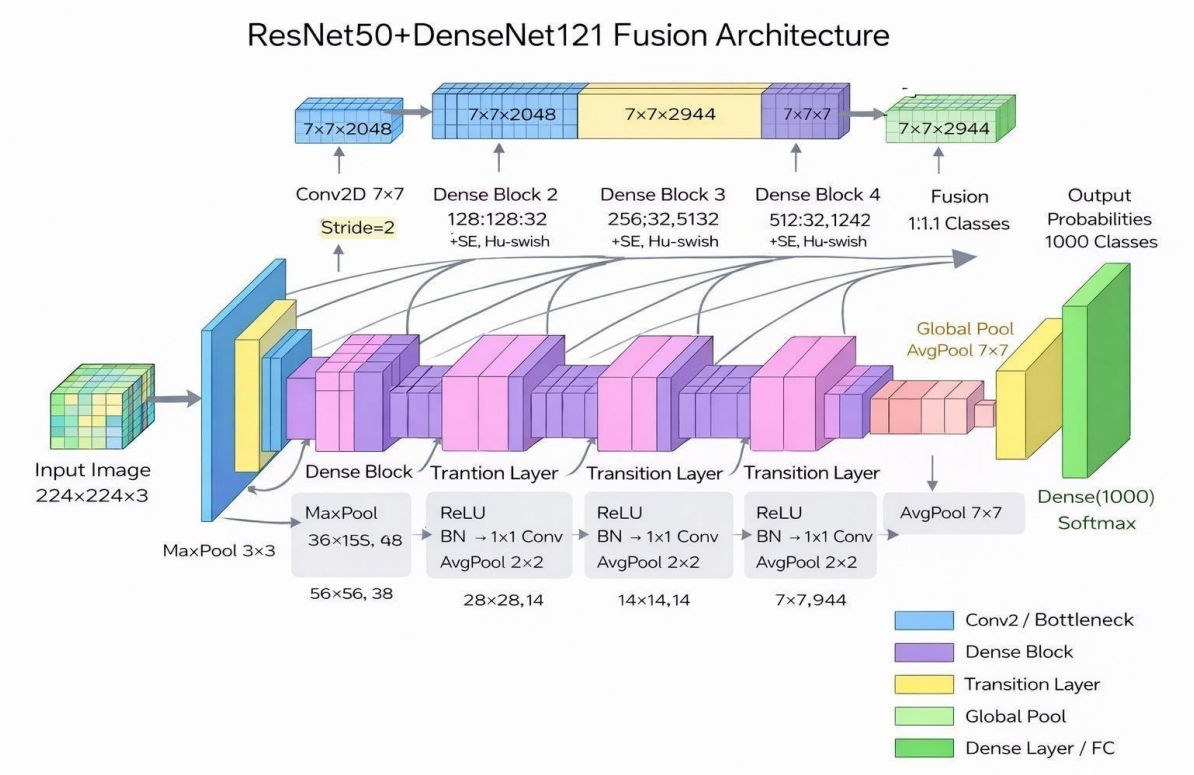


Figure 3.6: Best Model Architecture for Classification

3.5.1 Applied models for detection and Segmentation

3.5.1.1 YOLOv8-seg

For this research, we applied the YOLOv8 segmentation model from Ultralytics to automatically detect and mark the diseases in leaf and fruit images. We applied YOLOv8-Seg, which allows it to be performed in a single step and is capable of locating the infected areas, identifying what they are, and generating detailed pixel-level masks simultaneously. Unlike other object detection models that can only mark the affected areas with bounding boxes, YOLOv8-seg is capable of generating detailed segmentation masks. These masks are helpful in marking the specific infected areas, which is crucial for the accurate identification of disease spread and severity. Compared to two-stage models such as Mask R-CNN, YOLOv8-seg is faster and less computational, making it more suitable for real-time applications in agriculture. Other studies have also shown accurate performance using YOLO-based models for plant disease detection and segmentation, then validating their appropriateness for precision agriculture applications. YOLOv8-seg seeks to optimize a multi-task objective that combines detection and segmentation losses:

$$L_total = \lambda_1 L_box + \lambda_2 L_cls + \lambda_3 L_DFL + \lambda_4 L_mask \quad \text{.....(7)}$$

Parameter optimization follows gradient-based learning:

$\theta_{(}t+1_{)} = \theta_{t} - \eta \nabla_\theta L_total$ .....(8)

3.5.2 Architecture of YOLOV8-seg

There is a backbone for feature extraction, a neck for feature fusion at different scales, and parallel heads for detection and segmentation. The backbone identifies significant visual features such as color variation, lesion texture, and irregularities in the boundaries. The neck is responsible for the fusion of features at different scales to enable the detection of both small and large infected regions.The detection head predicts bounding boxes, class probabilities, and confidence scores. The segmentation head makes prototype masks and instance-specific coefficients to make accurate pixel-level disease masks. YOLOv8-seg improves a multi-task goal that includes losses for localization, classification, and segmentation. We employ IoU-based loss for learning bounding box regression, Binary CrossEntropy loss for learning classification, Distribution Focal Loss for learning localization refinement, and pixelwise mask loss for learning segmentation. Transfer learning was employed to facilitate faster convergence and generalization by initializing the model with pretrained weights. The model was trained for 100 epochs with 16 samples per batch and an input size of 640×640 pixels. To ensure robustness of the system in different field conditions, automatic data augmentation methods such as mosaic augmentation, random flipping, scaling, and HSV color space transformation were employed

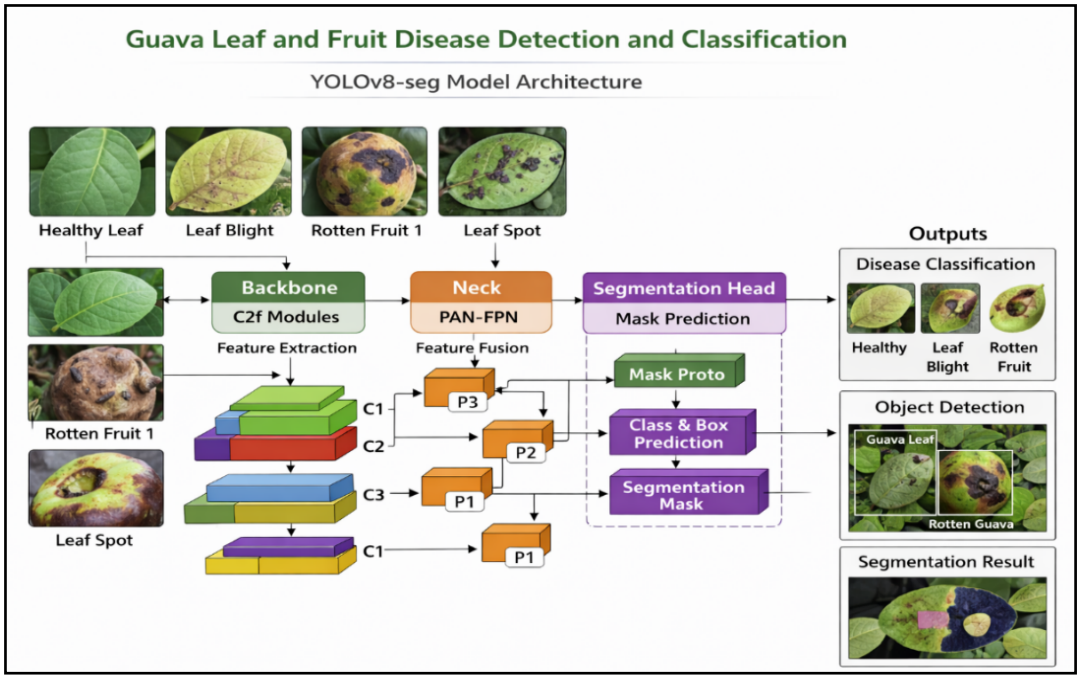


Figure 3.7: Best Model Architecture for Detection

3.5.1.2 Mask R-CNN model

In this study, Mask R-CNN was used to identify and then explain the diseases in guava leaves and fruits at both the object-instance and pixel-levels. This enabled the detection, categorization, and prediction of the exact mask of the object simultaneously[52]. The R50-FPN variant employs a ResNet-50 backbone for hierarchical feature extraction, Feature Pyramid Network (FPN) for multi-scale representation, Region Proposal Network (RPN) for proposing candidate ROIs, and ROIAlign for accurate spatial alignment. The network is divided into three parallel heads for mask prediction, bounding box regression, and classification. We began the model with weights pretrained on COCO and fine-tuned it on five disease classes using GPU training with the SGD optimizer and a tremendous amount of data augmentation (approximately 7,000 images).
The multi-task objective function optimized by Mask R-CNN is given by

$$L_{total} = L_{cls} + L_{box} + L_{mask} \quad \text{……. (9)}$$

The performance of the segmentation task was measured using the Intersection over Union (IoU) and Dice coefficient, given by

$$IoU = | M_{pred} \cap M_{gt} | / | M_{pred} \cup M_{gt} | \quad \text{…… (10)}$$

$$Dice = 2 | M_{pred} \cap M_{gt} | / (| M_{pred} | + | M_{gt} |) \quad \text{…….(11)}$$

Mask R-CNN consumes more computational power compared to single-stage detection architectures. However, it is one of the best architectures for object detection and fine-grained segmentation. It is a good method to monitor crop diseases and determine how serious they are.

Chapter 4 | Result and Performance Analysis

4.1 Overview

The ResNet50 + DenseNet121 Fusion model performed better than all other architectures. It had the best overall accuracy of 98.2%, the lowest test loss of 0.055, and had good generalization skills with balanced precision, recall, and F1-score values for all guava leaf and fruit disease classes. The ResNet50 + DenseNet121 Fusion model performed better than all other models that were tested. It had the best overall accuracy of 98.2%, the lowest test loss of 0.055, and it generalized well with all precision, recall, and F1-score values well-balanced for all guava leaf and fruit disease classes. The DenseNet121 and MobileNetV2 models also generalized well, but there were some mistakes in classes that looked similar to each other, such as Phytophthora and Styler & Root. The ROC and PR curve evaluation tests proved that the Fusion model was excellent at distinguishing classes. The PCA plot proved that most classes were well-clustered. The only overlapping classes were Styler & Root and Phytophthora. Since Mask R-CNN was more accurate and better at dealing with backgrounds, it performed better at detecting and distinguishing objects. However, YOLOv8-seg performed better at classes that were harder to detect and at recall. YOLOv8-seg has a more stable detection performance for different confidence levels, particularly for hard classes, as indicated by the F1-confidence curves and mAP results. These results show that choosing a suitable model and leveraging features can enhance the classification and detection of guava diseases.

4.2 Performance Evaluation for Deep Learning Models Classification.

We employed DenseNet121, NASNetMobile, MobileNetV2, InceptionV3, Exception, ResNet50+DenseNet121 Fusion, and EfficientDualBlock CNN to determine the diseases present in guava fruit and leaves because they are excellent at detecting features. We analyzed the performance of the model using the Confusion Matrix, ROC Curve, Loss Curve, and Accuracy Curve. These methods help us determine the accuracy of predictions, the ability of the model to distinguish between classes, the effectiveness of the training process, and the robustness of the model.

Confusion matrices for the deep learning-based classification models

The ResNet50 + DenseNet121 Fusion model was the most accurate overall, with 109 correct predictions out of 111 and only 2 wrong ones. Phytophthora and Styler & Root made the most mistakes, with one sample going each way. It was 100% accurate to find Disease Free, Red Rust, and Scab. Phytophthora had a recall rate of 95.83%, which means that it correctly identified 23 cases, missed 1 case, and falsely identified 1 case. Styler & Root got 95% of the answers right, with 19 true positives, 1 false negative, and 1 false positive. The F1 and precision scores were also very high (Phytophthora: 95.83%, Styler & Root: 95.00%), which means that there was very little confusion and the disease detection was very reliable. DenseNet121 got 97.30% of the answers right, with 108 correct and 3 wrong. Some of the wrong classifications were Phytophthora → Scab, Phytophthora → Styler & Root, and Styler

& Root → Phytophthora. There were no problems finding Disease Free, Red Rust, and Scab. It was 91.67% accurate for Phytophthora, 95.65% accurate for precision, and 93.62% accurate for the F1-score. Styler & Root maintained its 95% recall and precision, with an F1-score of 95%. DenseNet121 worked really well overall, but it made a few more mistakes in important disease classes. MobileNetV2 made 106 correct predictions and 5 wrong ones, for a total of 95.50%. Most of the errors were in Phytophthora. It was confused with Styler & Root twice and Scab once, and one Styler & Root was wrongly labeled as Phytophthora. There was no doubt that Disease Free and Red Rust were both found. Phytophthora's F1 score was 89.36%, its recall was 87.50%, and its precision was 91.30%. Styler & Root had a recall rate of 95%, a precision rate of 90.48%, and an F1-score of 92.68%. The model works fine, but Phytophthora gets very confused by it.

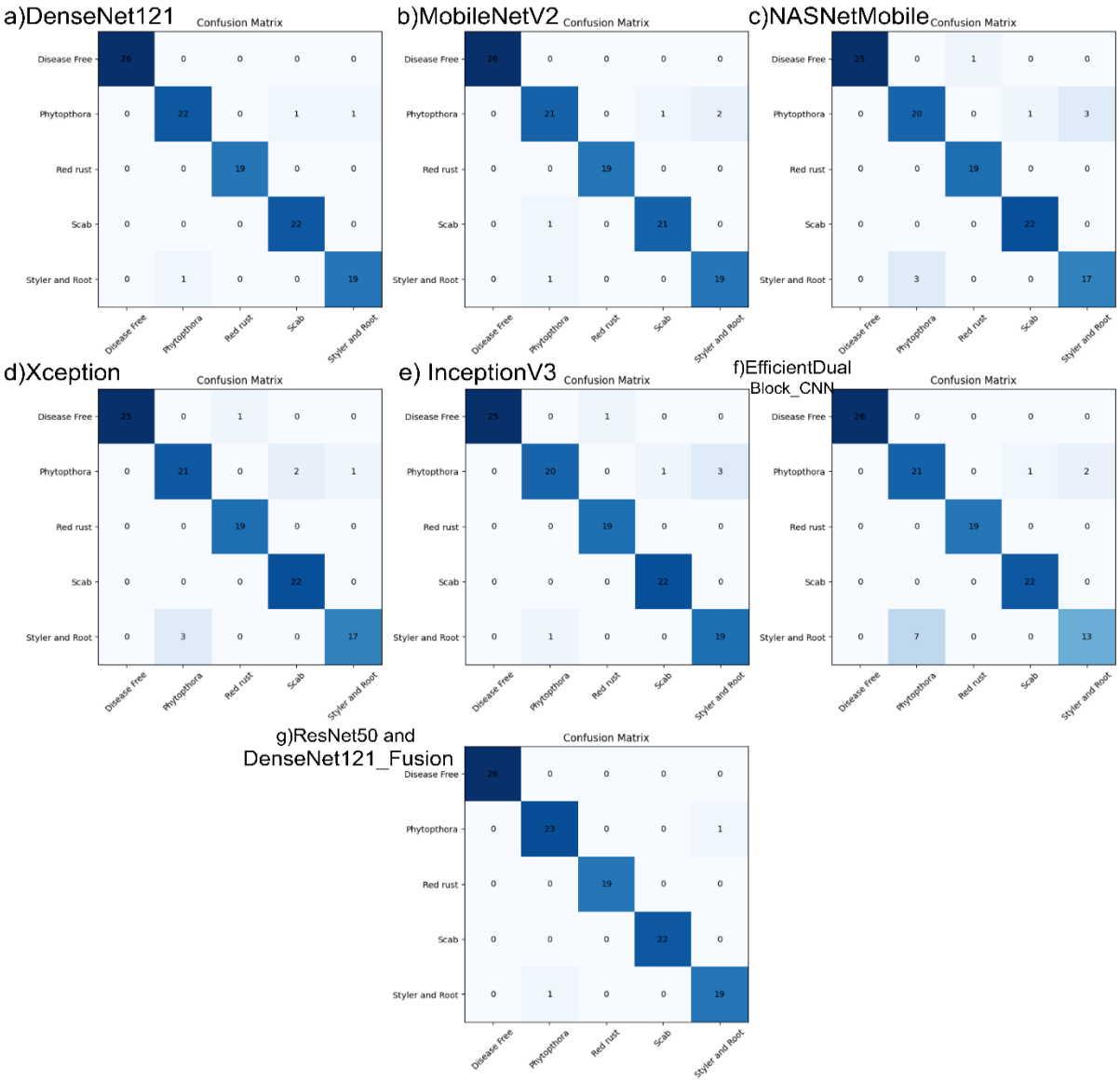


Figure 4.1: Confusion matrices for the deep learning-based classification models

InceptionV3 got 105 answers right and 6 wrong, for a total of 94.59%. Phytophthora was wrongly called Styler & Root three times, Scab once, Phytophthora once, and Disease Free once. The recall for Phytophthora dropped to 83.33%, the precision to 95.24%, and the F1-score to 88.89%. Styler & Root's recall stayed at 95%, its precision stayed at 86.36%, and its F1-score stayed at 90.48%. The model is pretty accurate, but it gets more confused in important classes. Xception got 104 answers right and 7 wrong, which is 93.69% of the time. There were big mistakes in Styler & Root → Phytophthora (3 times), Phytophthora → Scab (2 times), Phytophthora → Styler & Root (1 time), and Disease Free → Red Rust (1 time). Phytophthora had a recall and precision of 87.50% (F1-score: 87.50%), but Styler & Root's recall fell to 85%, precision to 94.44%, and F1-score to 89.47%. The number of times Disease Free, Red Rust, and Scab were found stayed high. The model is pretty close, but Styler & Root is the class that gets the most mixed up.

Table 4.1: Deep Learning Models Classification Table

Model	Accuracy %	Precision %	Recall %	F1 %	MCC %	PR AUC %	Test Loss %	Test Accuracy %
ResNet50DenseNet121_Fusion	98.20	98.17	98.17	98.17	97.74	99.75	5.55	98.20
densenet121	97.30	97.26	97.33	97.28	96.62	99.69	6.18	97.30
mobilenetv2	95.50	95.45	95.59	95.50	94.36	98.70	11.90	95.50
inceptionv3	94.59	94.45	94.90	94.52	93.31	98.72	12.16	94.59
xception	93.69	93.72	93.73	93.62	92.14	99.37	12.58	93.69
nasnetmobile	92.79	92.52	92.90	92.67	90.99	96.76	20.17	92.79
EfficientDualBlock_CNN	90.99	91.46	90.50	90.57	88.87	94.69	33.71	90.99

The model is quite accurate, but Styler & Root is the class that gets the most confused. NASNetMobile was able to get 92.79% of the answers correct (103 out of 103). The errors were mostly between Phytophthora and Styler & Root (3 each) and between Disease Free and Red Rust (1). Phytophthora had an F1-score of 85.11%, a recall of 83.33%, and a precision of 87%. Styler & Root had an F1-score of 85%, a recall of 85%, and a precision of 85%. The accuracy of disease-free, red rust, and scab detection was still high, but not as reliable because there was a lot of confusion between the critical classes of diseases. The poorest performance was given by EfficientDualBlock_CNN, which answered 90.99% of the questions correctly (101 correct and 10 incorrect). The main problem was Styler & Root → Phytophthora (7 cases). The F1 score was 80.77%, the recall was 87.50%, and the precision was 75%. The recall for Styler & Root went down to 65%, the precision to 86.67%, and the F1-score to

74.29%. It was still possible to find disease-free, red rust, and scab. This model had the most trouble finding important classes, especially Styler & Root.

4.2.1 ROC curves and AUC Analysis

ResNet50 + DenseNet121 Fusion, the winning model, achieved the highest overall AUC of 0.999+. The class-wise AUC was perfect (1.0) for Disease Free, Red Rust, and Scab, and near-perfect for Phytophthora (0.999) and Styler & Root (0.998). The ROC curves were practically hugging the top-left corner for all classes, further affirming minimal FP, high TPR, and outstanding discriminative power relative to all single-backbone architectures.

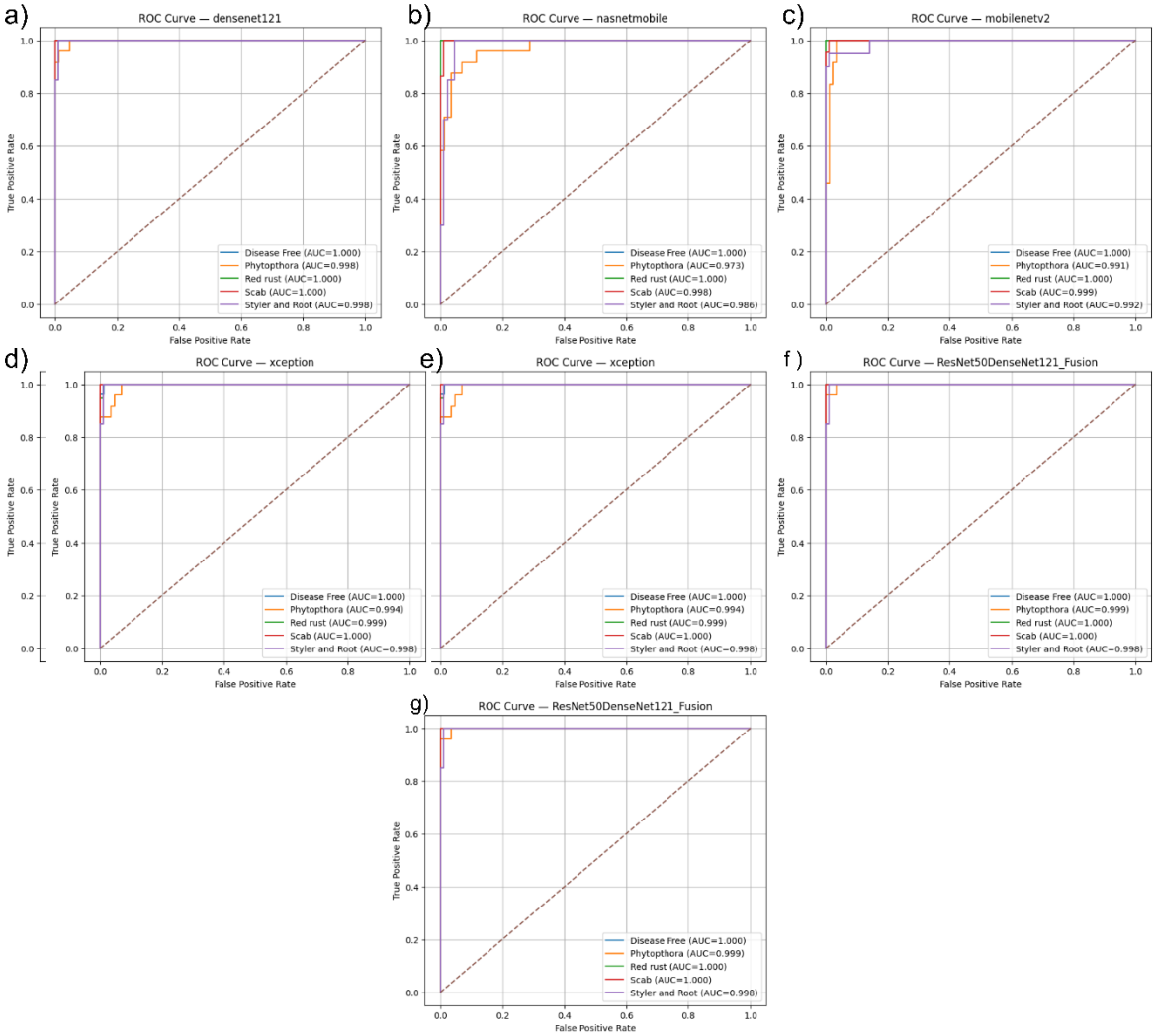


Figure 4.2: ROC curves of the evaluated deep learning classification models

DenseNet121 achieved near-perfect discrimination with an overall AUC of approximately 0.999. Class wise AUC was 1.0 for Disease Free, Red Rust, and Scab, and 0.998 for Phytophthora and Styler & Root. The curves hugged the top-left corner closely, indicating

very high true positive rate (TPR) and near-zero false positive rate (FPR). Minor deviations were observed in the cases of Phytophthora and Styler and Root, yet the model maintained excellent robustness. MobileNetV2 resulted in an overall AUC of 0.996, with class-wise perfect or near- perfect AUCs (Phytophthora 0.991, Styler & Root 0.992). The ROC curves were veryclose to ideal curves, although not as smoothly as DenseNet121 or InceptionV3, indicating a slight degradation of performance for more difficult classes. InceptionV3 performed well with an average AUC of 0.996-0.998. Although disease-free, red rust and scab achieved a perfect AUC, Phytophthora and styler and root were slightly lower at 0.992. The ROC curve was close to the perfect upper-left corner, indicating little class-specific confusion but overall strong multi-class discrimination. Xception achieved an overall AUC of 0.997–0.998. While Disease Free, Scab, and Red Rust had near perfect separation, Phytophthora (0.994) and Styler & Root (0.998) showed minor reduction. The model was highly competitive, slightly below Fusion and DenseNet121. NASNetMobile resulted in an overall AUC of 0.991-0.993. Phytophthora AUC decreased to 0.973, showing a clear confusion with other classes. Styler & Root was also slightly impacted (0.986). The ROC curve was less steep, reflecting a less confident separation for the most difficult disease classes. EfficientDualBlock_CNN, the weakest performing model in ROC analysis, had an overall AUC of 0.975– 0.98. Disease Free, Red Rust, and Scab retained complete separability, but Phytophthora (0.95) and Styler & Root (0.972) were less discriminative. The ROC curves were less smooth, indicating confusions between these two important classes.

Overall, the comparative ROC analysis confirms that feature fusion significantly enhances classification reliability. While all pre-trained architectures demonstrated strong performance, the Fusion model outperformed individual backbones in terms of stability, robustness, and class-wise discrimination. Additionally, across all models, Phytophthora and Styler & Root remained the most challenging classes due to their visual similarity, whereas Disease Free and Red Rust were consistently well separated.

4.2.2 Learning loss and accuracy curves Analysis

The analysis of the accuracy and loss curves side by side shows that the models being tested learn in different ways. The best model was ResNet50 + DenseNet121 Fusion because it converged the fastest, reaching about 99.8% training accuracy and had high validation and test accuracy (about 98%). The training and validation loss decreased smoothly, with only a small variation in change (training loss ≈0.01, validation loss ≈0.06-0.08, test loss ≈0.055). This shows that the model works well most of the time and that the optimization algorithm is very stable. DenseNet121 had a strong baseline with near-perfect training accuracy (around 100%), high validation accuracy (around 98%), and test accuracy (around 97%), very low loss values (training loss ≈0.01, validation loss ≈0.05-0.07, test loss ≈0.065), and a small generalization gap. MobileNetV2, as a light model, had training accuracy of around 100% and test accuracy of 95-96%. Its loss values were stable (validation loss 0.05-0.07, test loss ≈0.12), showing that it was well-trained and converged smoothly (7-9 epochs). InceptionV3 had high

accuracy in classifying 99% of the training data and 94-96% of the test data. It had a small amount of validation oscillations (loss 0.07-0.13), showing a slight overfitting but smooth learning. Xception learned quickly and got almost everything right on the training set (about 100%) and the test set (about 93–94%). It also had a moderate validation oscillation (loss 0.08–0.12), which meant that it was good at finding features without overfitting too much. NASNetMobile got almost all of the training right (99%), but the validation accuracy was very unstable (90–98%) and the test loss was higher (about 0.20), which showed overfitting and less stable optimization. The model that didn't work well, EfficientDualBlock_CNN, had unstable learning, big changes in validation (80–94%), a big difference between training and validation, and a high final validation loss (~0.88). This made it difficult to apply what we learned to other problems and resulted in the lowest test score (approximately 91%). Models such as Fusion and DenseNet121, which had less variation in validation and a smaller generalization gap converged better, optimized better, and performed better on tests. However, models such as EfficientDualBlock_CNN were not as reliable since they were not stable and overfitted.

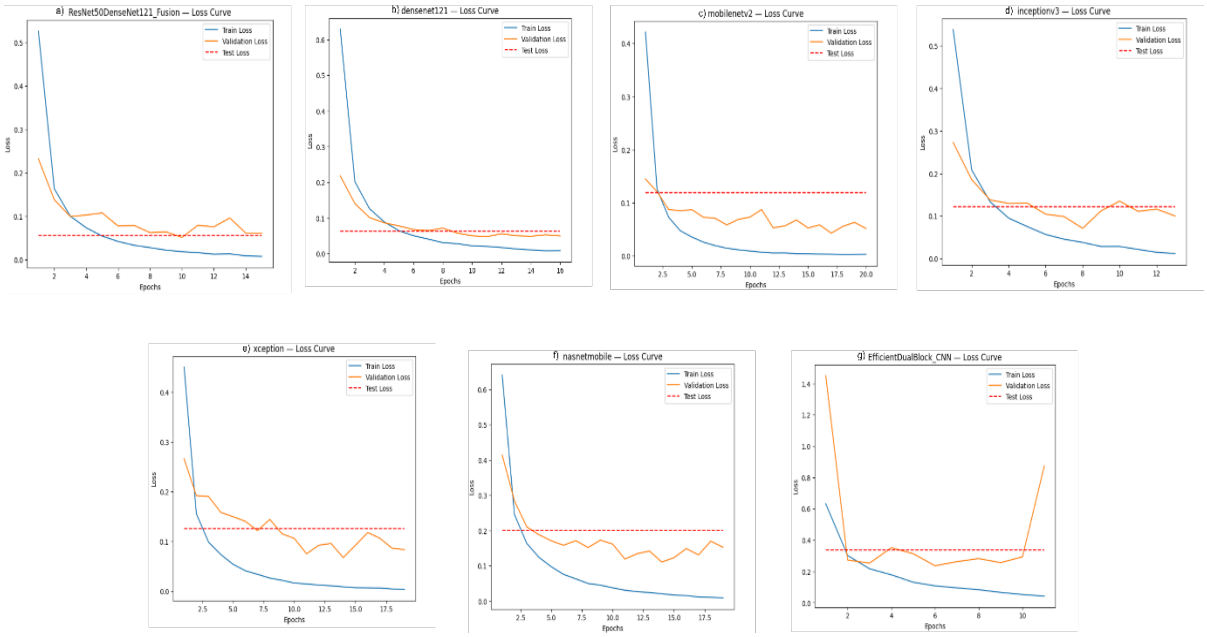


Figure 4.3: Learning loss curves over epochs of the tested deep learning classification models

4.2.3 Overall numbers for all the models' classifications

The ResNet50 + DenseNet121 Fusion model did the best overall, getting 98.20% of its predictions right (109 out of 111). The F1-score was 98.17%, which means that there were as many false positives as true positives. This shows that the two were very well balanced. The MCC of 0.9774 shows that the predicted and real labels are very close to each other in all five classes. The ROC-AUC values for each class (0.998–1.000) show that the classes can be

almost perfectly separated. The PR-AUC value of 0.9975 shows that the precision-recall performance is still good even though the classes aren't the same. The model had the lowest test loss (0.0555), and the learning curves showed that it learned quickly with a small difference between training and validation. This means that it didn't fit too well and worked well overall. DenseNet121 was 97.30% accurate overall. It had an F1-score of 97.28%, a recall of 97.33%, and an accuracy of 97.26%. It always did well in all of its classes.

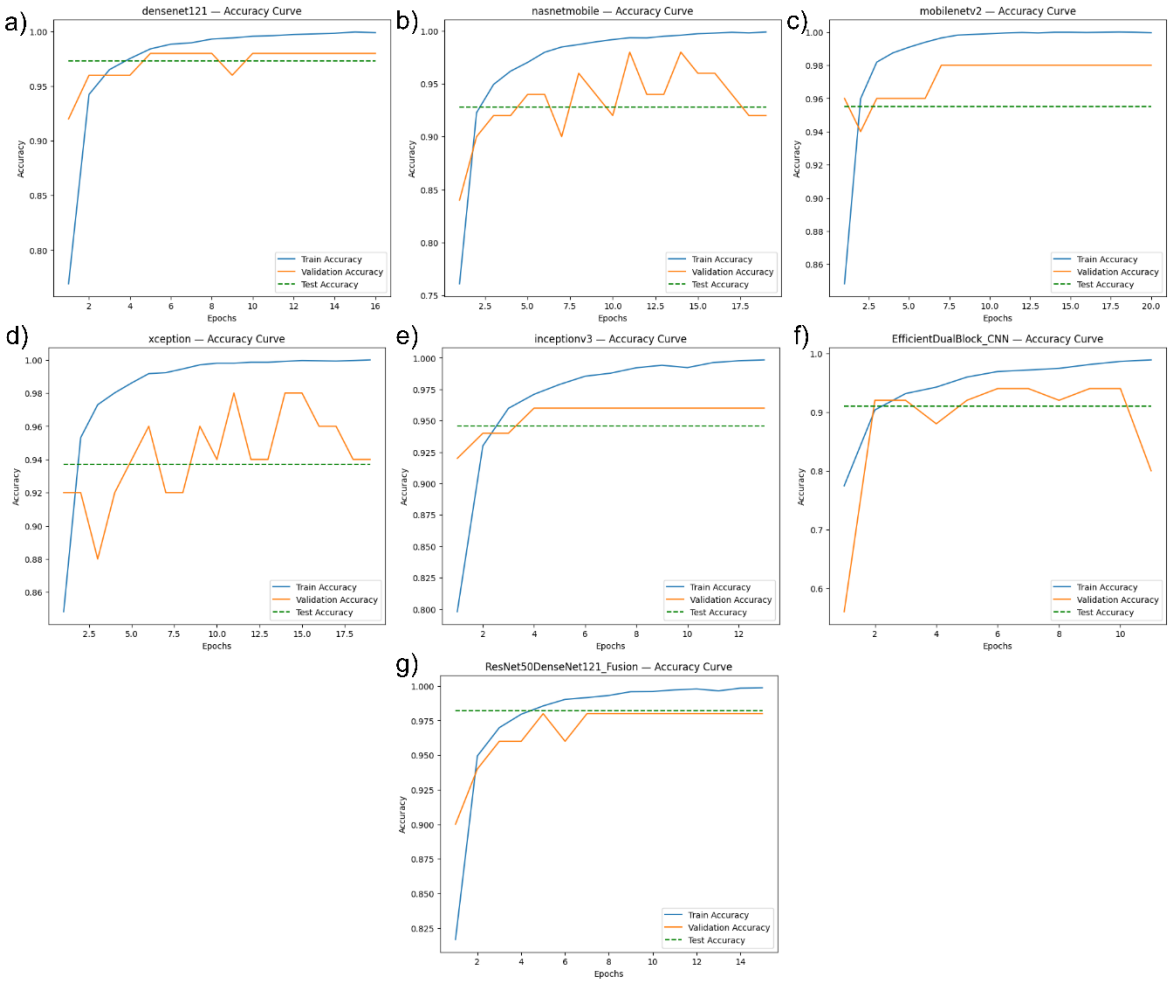


Figure 4.4: Learning accuracy curves over epochs of the tested deep learning classification models

A MCC of 0.9662 means that the model is very good at making predictions for more than one class. The ROC-AUC values, which range from 0.998 to 1.000, show that the classes are very different from each other. The PR-AUC value (0.9969) shows that the behavior of precision and recall is the same. DenseNet121 had a low test loss (0.0618) and smooth training-validation curves, which showed that it was very good at generalizing. It wasn't as stable as the Fusion model, but it still did a great job of telling diseases apart. MobileNetV2 had an accuracy of 95.50%, with the same levels of precision (95.45%), recall (95.59%), and F1-score (95.50%). The MCC of 0.9436 shows that there is a strong link between predictions and

true labels, but it could be better with more layers. The ROC-AUC values (0.991–1.000) show that the classes are very different from each other, and the PR-AUC value (0.9870) shows that the balance between recall and precision is stable. The test loss was moderate (0.1190), and the learning curves leveled off smoothly, with a small gap between training and testing. This means that the model can still do a good job in many situations, even though it is small and doesn't need a lot of computing power. InceptionV3 had a 94.59% accuracy rate, a 94.45% precision rate, a 94.90% recall rate, and a 94.52% F1-score. This shows that it works well with more than one group. The 0.9331 MCC shows that the guesses are correct. The ROC-AUC values (0.992–1.000) show that the classes are very different from each other, and the PR-AUC value (0.9872) shows that the precision-recall's performance is stable. The test loss was a little higher than the training loss, at 0.1216. The learning curves showed that the model was stable and had a small generalization gap. This means that the model fit a little too well.

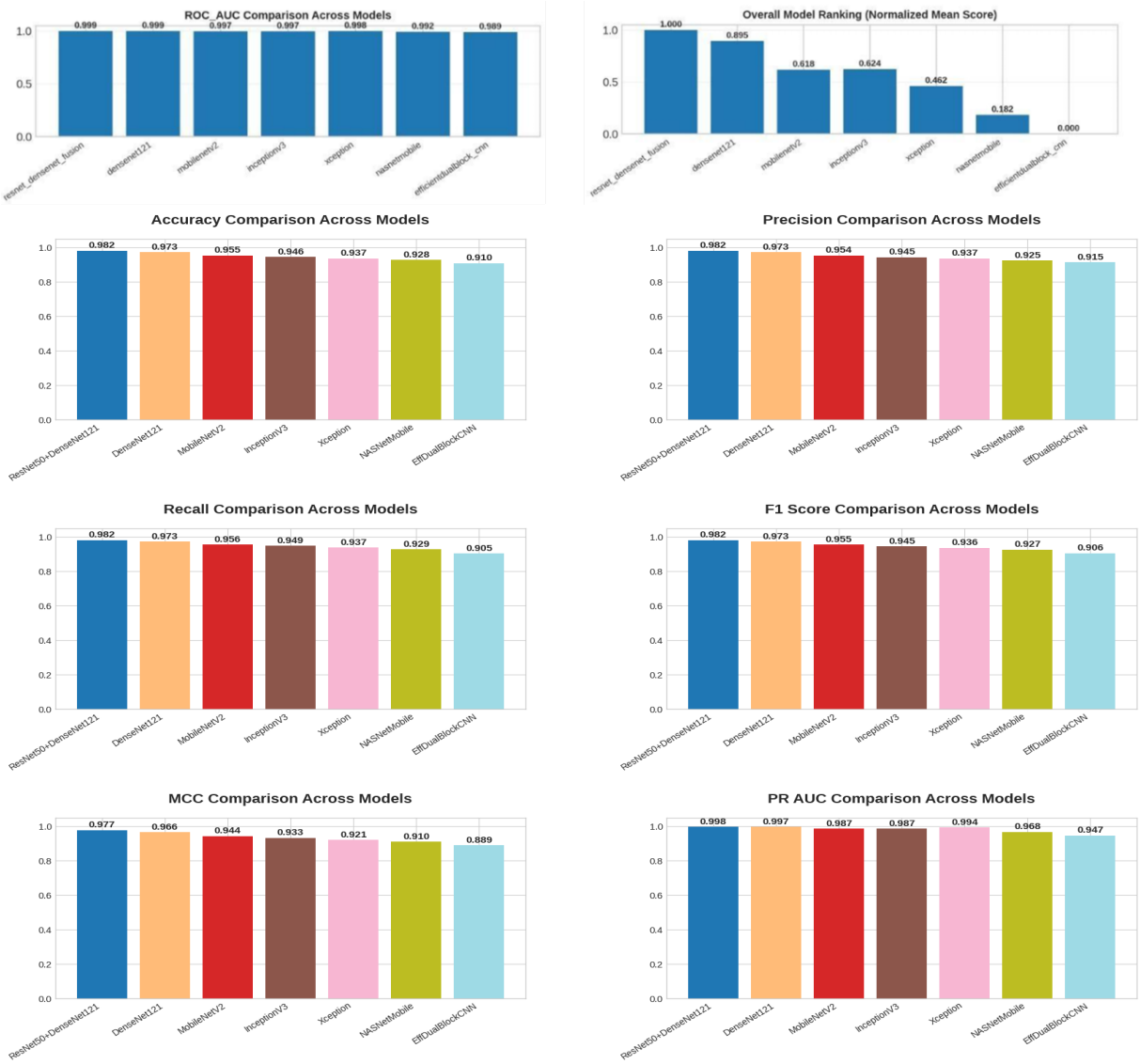


Figure 4.5: Overall Classification Metrics Comparison Across All Models

Table 4.2: Performance Comparison Across All using Model

Model	ResNet50 + DenseNet121 Fusion	DenseNet121	MobileNetV2	InceptionV3	Xception	NASNetMobile	EfficientDualBlock_CNN
Accuracy (%)	98.20	97.30	95.50	94.59	93.69	92.79	90.99
Precision (%)	98.17	97.26	95.45	94.45	93.72	92.52	91.46
Recall (%)	98.17	97.33	95.59	94.90	93.73	92.90	90.50
F1 (%)	98.17	97.28	95.50	94.52	93.62	92.67	90.57
MCC	0.9774	0.9662	0.9436	0.9331	0.9214	0.9099	0.8887
ROC-AUC (Range)	0.998–1.000	0.998–1.000	0.991–1.000	0.992–1.000	0.994–1.000	0.973–1.000	0.950–1.000
PR-AUC	0.9975	0.9969	0.9870	0.9872	0.9937	0.9676	0.9469
Test Loss	0.0555	0.0618	0.1190	0.1216	0.1258	0.2017	0.3371
Generalization Stability	Excellent	Very Strong	Strong	Stable (Mild Overfitting)	Moderate	Moderate–Weak	Weak (Overfitting)

The overall accuracy of Xception was 93.69%, with balanced precision (93.72%), recall (93.73%), and F1-score (93.62%). The MCC of 0.9214 shows that the predictions are right, but it's not as high as the MCCs of other deep networks. ROC-AUC values (0.994–1.000) are still very close to perfect, but the overall accuracy went down a little because of some small mistakes. The PR-AUC (0.9937) shows that the way precision and recall work together is very good. The test loss was moderate (0.1258), and the learning curves showed that the two were getting closer. There was a difference between training and testing. This implies that the stability of the generalization was also moderate. It was correct 92.79% of the time. It had an accuracy of 92.52%, a recall of 92.90%, and an F1 score of 92.67%. The multi-class predictive strength is okay, not great, with an MCC of 0.9099. The ROC-AUC scores (0.973-1.000) indicate that the disease classes that appear similar are not as different as they appear. The PR-AUC score (0.9676) indicates that the precision-recall reliability is not as good. The test loss was higher (0.2017), and the validation showed that the stability of the generalization was moderate to weak, especially for disease classes that were difficult to understand. The custom

EfficientDualBlock_CNN had the lowest accuracy, which was 90.99%. It had a precision of 91.46%, a recall of 90.50%, and an F1 score of 90.57%. This implies that the predictions were not as even. This is the worst result for the task of predicting multiple classes, with an MCC of 0.8887. The ROC AUC values (0.950-1.000) and the PR AUC (0.9469) indicate that the tests did not perform well in distinguishing between Phytophthora and Styler & Root. The worst loss on the test was 0.3371. The learning curves indicated that the validation loss increased and the validation accuracy improved over time. This indicated that the model was not performing well on new data and was too specific.

4.3 Extra Analysis for Classification

4.3.1 McNemar's and Chi-square test:

McNemar's test is required in this study because it allows us to compare the performance of paired classification models on the same test data in a statistical manner, rather than simply comparing the accuracy of the models. We tested all of the models on the same 111 test images for each sample, so we can easily compare the results. The McNemar test shows you how often one model is right when the other is wrong. This allows you to not only compare the accuracy of the error, but also how it is distributed. When comparing to ResNet50+DenseNet121 Fusion as a baseline, only the comparison with EfficientDualBlock_CNN is significant ($p = 0.0433$, significant at $p<0.05$). The Fusion model was better at predicting things, and it made different kinds of errors. The comparisons to NASNetMobile ($p = 0.1138$), Xception ($p = 0.1824$), MobileNetV2 ($p = 0.2482$), InceptionV3 ($p = 0.2898$), and DenseNet121 ($p = 1.0000$) are not significant at 0.05. This means that the error distribution of the fusion model is not always different from the baselines with this sample size. The only one that is different enough to be noticed is EfficientDualBlock_CNN.

Table 4.3.1: Pairwise Comparisons: MobileNetV3Small vs All Other Models

Pairwise Comparisons: ResNet50DenseNet121_Fusion vs All Other Models

Reference Model	Compared Model	Better Model	McNemar p-value	Significant (p<0.05)	Chi-square p-value
ResNet50DenseNet121_Fusion	EfficientDualBlock_CNN	ResNet50DenseNet121_Fusion	0.0433	Yes	0.0209
ResNet50DenseNet121_Fusion	nasnetmobile	ResNet50DenseNet121_Fusion	0.1138	No	0.0578
ResNet50DenseNet121_Fusion	xception	ResNet50DenseNet121_Fusion	0.1824	No	0.0956
ResNet50DenseNet121_Fusion	mobilenetv2	ResNet50DenseNet121_Fusion	0.2482	No	0.0833
ResNet50DenseNet121_Fusion	inceptionv3	ResNet50DenseNet121_Fusion	0.2888	No	0.1573
ResNet50DenseNet121_Fusion	densenet121	ResNet50DenseNet121_Fusion	1.0000	No	0.3173

4.3.2 Anova Test

The ANOVA (Analysis of Variance) test was used on the ResNet50DenseNet121_Fusion model to determine which feature dimensions are the most discriminative for the five guava classes: Disease-Free, Phytophthora, Red Rust, Scab, and Styler & Root. The dependent variable is the feature value, and the independent variable is the class label. ANOVA is the correct test to use in this case because there are more than five groups being compared, and a t-test is only valid for two groups. The F-statistic is a measure of the ratio of The ratio of between-group variance to within-group variance, and the larger the F-score, the more separation between classes. In the results, the top discriminative features are feature index 222 (F=425.3), 176 (F=397.2), 256 (F=395.2), and 329 (F=386.0), and even lower-ranked features like feature 434 (F=235.2) are still significant. The large F-scores indicate that the p-values are << 0.05, which means that the differences between the class means are statistically significant. This shows that the Fusion model is very discriminative, and the top and lower-ranked features all contribute to the separation of classes.

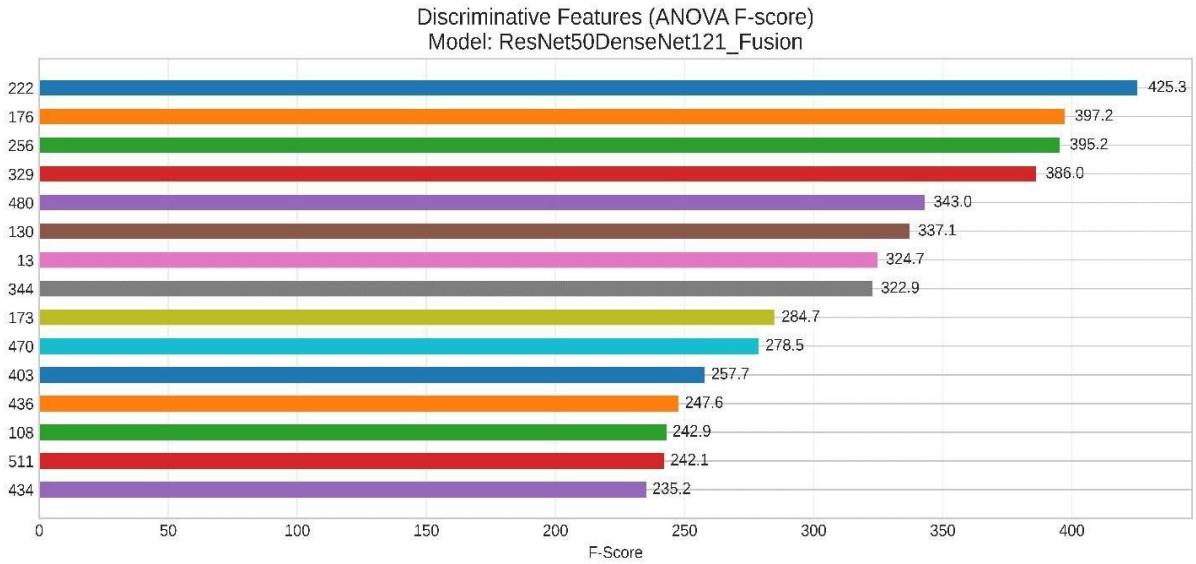


Figure 4.6: Discriminative Features (ANOVA F-score)

Table 4.4: Anova test

Feature	F-score	Significance ($p < 0.05$)
222	425.3	Extremely Highly significant
176	397.2	Highly significant
256	395.2	Highly significant
329	386.0	Highly significant
434	235.2	Significant

4.3.3 PCA Separability

PCA Analysis – ResNet50DenseNet121_Fusion Principal Component Analysis (PCA) was performed on the deep features with high dimensionality (≈3072 from the concatenation of ResNet50 penultimate features ≈2048 and DenseNet121 ≈1024) to visualize the feature space and determine the separability of classes in the learned feature space. The first two principal components (PC1 = 29.3%, PC2 = 16.4%) explain 45.77% of the total variance, which is a significant low-dimensional representation of the embedding's.

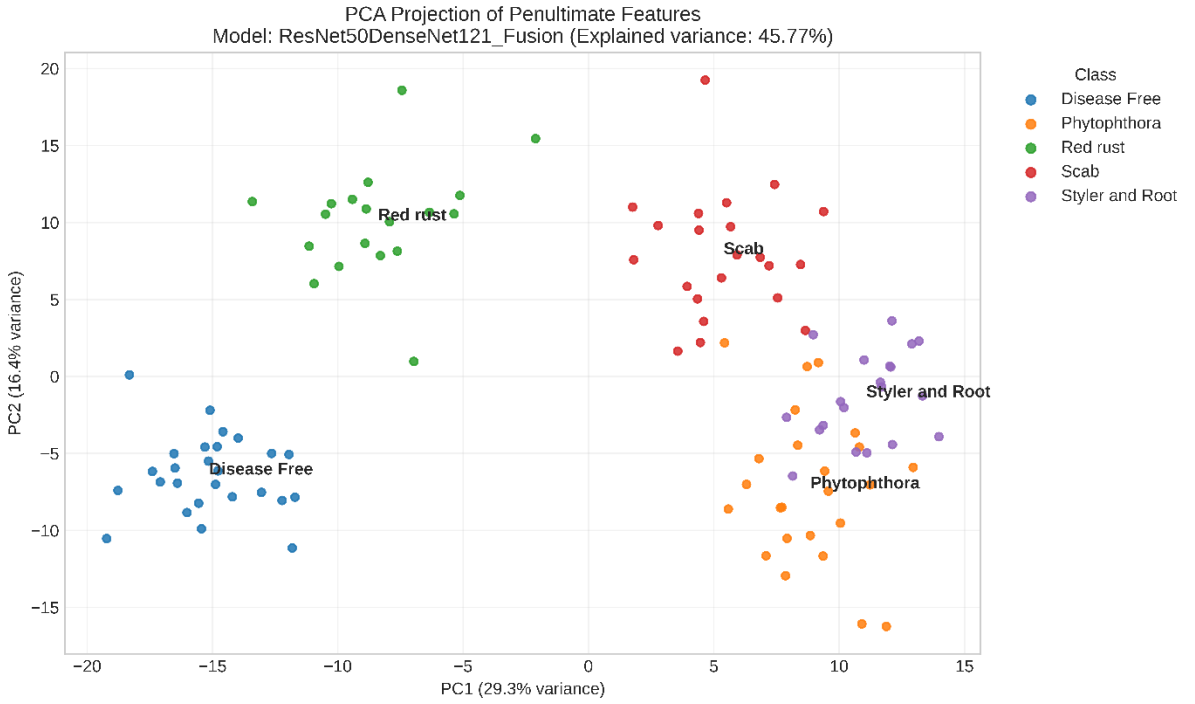


Figure 4.7: PCA Projection of Penultimate Features

The projected feature space reveals that Disease-free is a separate cluster entirely distinct from other classes, Red Rust is a separate cluster distinctly above Disease-Free, and Scab is a compact cluster in the mid-right position. Phytophthora and Styler & Root are located close together in the bottom-right corner, overlapping partially, which explains their higher confusion in classification. In general, the PCA verifies that the Fusion model has learned discriminative features well, since most classes are well-separated in the 2D feature space, except for Phytophthora and Styler & Root, which have some overlap. This indicates that the ResNet50DenseNet121_Fusion model has learned robust embeddings that can be used to distinguish disease classes effectively.

Performance Comparison:

The performance of seven deep learning models: ResNet50 + DenseNet121 Fusion, DenseNet121, MobileNetV2, InceptionV3, Xception, NASNetMobile, and EfficientDualBlock CNN, was compared on the basis of three important parameters:

Balanced Accuracy, Macro Specificity, and Matthews Correlation Coefficient (MCC). Balanced Accuracy, which calculates the average of sensitivity and specificity, is a crucial parameter in case of imbalanced datasets, and the highest value of Balanced Accuracy was recorded by the Fusion model (0.9817), which clearly signifies that the classification is most balanced for all five classes of guavas, while the lowest value was recorded by EfficientDualBlock CNN Specificity is the ratio of correctly predicted negatives for all classes, and all models scored above 0.97, with the Fusion model scoring the highest at 0.9955, indicating the lowest number of false positives.

Performance Comparison: Balanced Accuracy, Specificity, and MCC

Model	Balanced Accuracy	Specificity (macro)	MCC
ResNet+DenseNet Fusion	0.9817	0.9955	0.9774
DenseNet121	0.9733	0.9933	0.9662
MobileNetV2	0.9559	0.9888	0.9436
InceptionV3	0.9490	0.9867	0.9331
Xception	0.9373	0.9842	0.9214
NASNetMobile	0.9290	0.9821	0.9099
EfficientDualBlock CNN	0.9050	0.9773	0.8887

Figure 4.8: Performance Comparison- Balanced Accuracy, Specificity, and MCC

The Matthews Correlation Coefficient (MCC), a measure that takes into account TP, TN, FP, and FN, is a balanced measure of prediction accuracy; the Fusion model scored the highest at 0.9774, indicating the best prediction consistency, with EfficientDualBlock CNN scoring the lowest at 0.8887. The ResNet50 + DenseNet121 Fusion model performed better than all other models on all measures, followed by DenseNet121 and MobileNetV2.

4.3.4 Detection & Segmentation Models Performance:

4.3.4.1 Confusion matrix Analysis for Yolov8-seg and mask r-cnn

The Mask R-CNN model was tested on 104 test images, resulting in an overall accuracy of 88.5% with a balanced accuracy of 0.88. The confusion matrix revealed high dominance along the diagonal, indicating high true positives and true negatives with low false positives and false negatives. On a per-class basis, Disease Free and Red Rust had perfect precision, recall, F1-score, IoU, and Dice (1.0). Phytophthora had moderate performance (precision 0.778, recall 0.913, F1-score 0.840, IoU 0.724), while Scab had perfect precision but slightly lower recall (F1-score 0.857, IoU 0.75). Styler & Root was the most difficult class, with the lowest precision (0.70), recall (0.737), F1-score (0.718), and IoU (0.56). The model had low false positive density (11.5%), high mean IoU (0.807), Dice (0.883), and pixel accuracy (88-90%).

Detection mAP@0.5 was 0.831 and mAP@0.5:0.95 was 0.675, suggesting strong performance even for separate classes.

The YOLOv8-seg model was evaluated on 114 samples, yielding a total accuracy of 66.7% based on the confusion matrix, with a balanced accuracy of 0.88-0.89. The class-wise accuracy revealed moderate diagonal dominance, where Disease Free had excellent detection (precision 0.828, recall 0.923, F1-score 0.873, IoU 0.774), while Phytophthora, Red Rust, and Scab had moderate detection with some degree of inter-class confusion. Hard Class Styler and Root had low precision (0.529) but excellent recall (0.900), F1- score 0.667, and IoU 0.50, indicating better hard class detection than Mask R-CNN. The false positive density rate was higher (24.6%), and background processing was relatively poor. The YOLOv8-seg model had better detection mAP@0.5 (0.907) and mAP@0.5:0.95 (0.783), as well as segmentation mAP@0.5 (0.889) and mAP@0.5:0.95 (0.769), performing better than Mask R-CNN at strict IoU conditions.

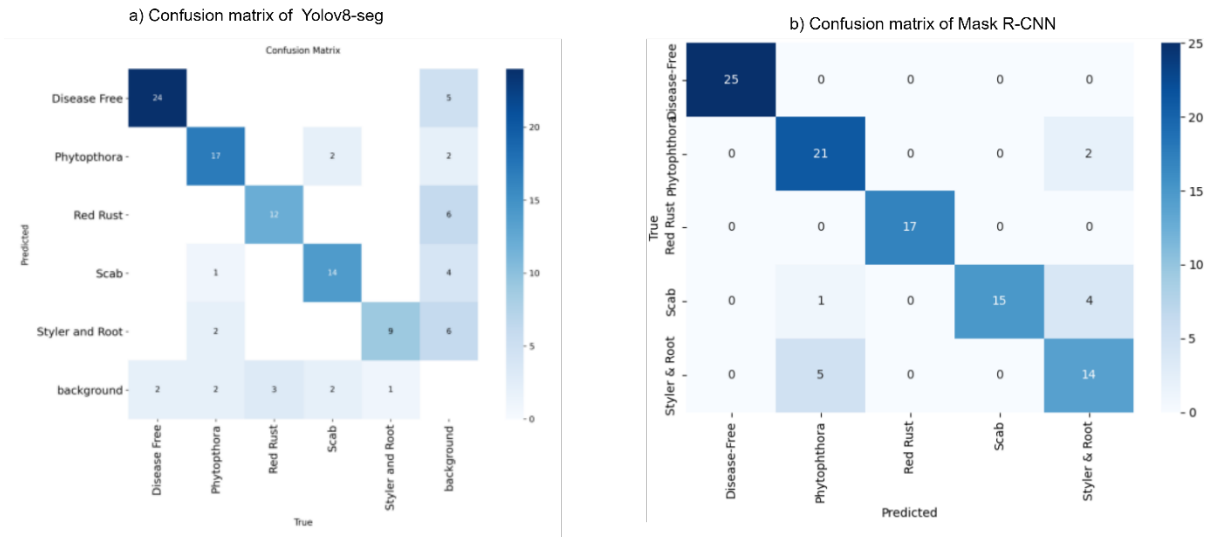


Figure 4.9: Confusion matrix for detection and segmentation

Table 4.5: Detection and Segmentation Metrics

Model	Task	mAP@0.5	mAP@0.5:0.95	Precision	Recall	F1-score
YOLOv8-seg	Detection (BBox)	0.907	0.783	0.899	0.881	0.890
Mask R-CNN	Detection (BBox)	0.831	0.675	0.900	0.587	0.711
YOLOv8-seg	Segmentation (Mask)	0.889	0.769	0.888	0.870	0.879
Mask R-CNN	Segmentation (Mask)	0.825	0.691	~0.896**	~0.880**	0.883**

Both models had a similar balanced accuracy value of approximately 0.88. Mask R-CNN had a higher precision value, lower FP density (11.5%), improved diagonal dominance for various classes, and more stable background processing. On the other hand, YOLOv8-seg had a higher

mAP@0.5 and mAP@0.5:0.95 value, slightly higher IoU and Dice values, and better hard-class recovery performance because of its higher recall rate. Thus, Mask R-CNN is more preferable if higher precision and lower false positives are given higher importance, while YOLOv8-seg has overall superiority.

4.3.4.2 The precision-recall (PR) curves Analysis

The YOLOv8-seg model was tested on the validation set at IoU = 0.5 using class-wise Precision-Recall (PR) curves, where Average Precision (AP) is evaluated as the area under each curve. Disease Free obtained the highest AP of 0.924, followed by Scab with 0.911 and Phytophthora with 0.887, which indicates excellent detection performance. Styler & Root obtained 0.849, while Red Rust obtained the lowest AP of 0.800, which indicates relatively higher detection difficulty. The overall detection performance was achieved at mAP@0.5 = 0.874, which indicates excellent average bounding box accuracy with a good precision-recall trade-off, where precision starts to decrease at higher recall regions because of higher false positives.

In the case of masked R-CNN, the class-based PR curve at IoU = 0.5 revealed that Disease Free and Scab had higher accuracy compared to broad recall, resulting in better AP scores, whereas the accuracy decreased rapidly for Phytophthora, Red Rust, and Styler and Root with high recall, indicating lower AP and higher false positives. The mAP@0.5 score indicates balanced multi-class identification, with easier identification for visually distinct classes and reduced PR performance for more difficult classes. Comparison between the models shows that higher and more stable PR curves result in larger areas under the curve and better mAP@0.5, indicating the role of class difficulty in the success and detection performance of Mask R-CNN.

While YOLOv8-Seg has higher AP scores for the styler and root and red rust difficult classes, Mask R-CNN has more accuracy for the apparently unique classes, indicating a trade-off between recall gain and memorability in multi-dimensional disorder processing

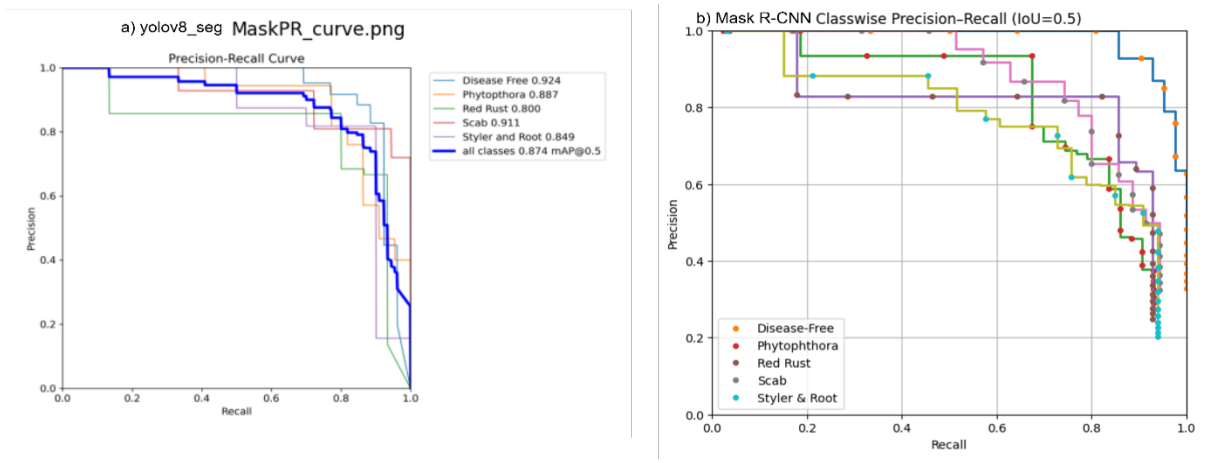


Figure 4.10: The precision-recall (PR)

4.3.4.3 F1-Confidence curve Analysis

The F1-confidence curves show the correlation between the confidence threshold and the harmonic mean of precision and recall for YOLOv8-seg and Mask R-CNN. YOLOv8-seg has the highest average F1 of around 0.89, at a confidence threshold of 0.65, while disease-free has the highest class-wise F1 peak (>0.9), while Styler & Root has a lower peak for F1 and greater sensitivity to the confidence there should. Phytophthora and Red Rust have peaks in the mid-confidence range, and Scab has stable performance before a slight drop-off at very high confidence values. The all-classes F1 score plot remains above 0.8 for a broad confidence range (~0.4-0.8), indicating excellent precision-recall trade-off.

For Mask R-CNN, there is a lower peak mean F1 of about 0.81 at the same confidence threshold of ~0.65. Disease Free has the highest class-wise F1, while Phytophthora and Red Rust have moderate gains at mid-level confidence thresholds. Styler & Root has the highest sensitivity, with steeper F1 drops at higher confidence values. The all-classes plot for Mask R-CNN has a smaller region of high F1, signifying slightly reduced robustness to different thresholds. In general, both models have a comparable optimal confidence threshold of approximately 0.65; however, YOLOv8-seg has a consistently higher and more stable F1 score over a broader range, indicating a better balance between precision and recall. This is in line with the mAP@0.5 and detection scores, and it is clear that YOLOv8-seg performs better than Mask R-CNN in terms of recall, hard-class detection, and overall detection performance. Therefore, YOLOv8-seg can be concluded to be the more efficient and reliable model for multi-class plant leaf disease detection in this research.

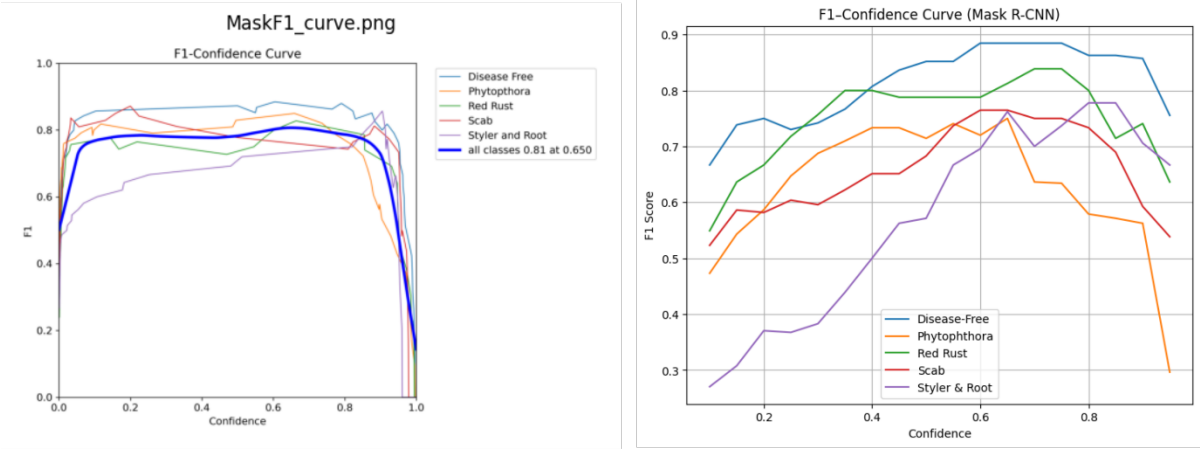


Figure 4.11: shows the F1-Confidence curve

Table 4.6 : Performance Comparison Across All using Model

Metric	Mask R-CNN	YOLOv8-seg	Observation
Test Samples	104	114	Different test splits
Overall Accuracy (CM)	88.5%	66.7%	Mask R-CNN higher (CM-based)
Balanced Accuracy	~0.88	~0.88–0.89	Nearly similar
Macro Precision	~0.896	~0.595	Mask R-CNN higher precision
Macro Recall	~0.88	~0.696	Mask R-CNN more balanced
Macro F1-score	~0.883	~0.636	Mask R-CNN higher (CM-based)
Total False Positives	12	28	YOLOv8 higher FP
FP Density	11.5%	24.6%	YOLOv8 more over-detection
Mean IoU (CM-based)	0.807	0.518	Mask R-CNN higher (CM)
Mean IoU (COCO est.)	0.807	0.82–0.85	YOLOv8 slightly higher
Mean Dice	0.883	0.88–0.90	Very close
Pixel Accuracy	88–90%	~88–90%	Comparable
Detection mAP@0.5	0.831	0.907	YOLOv8 superior
Detection mAP@0.5:0.95	0.675	0.783	YOLOv8 superior
Segmentation mAP@0.5	0.825	0.889	YOLOv8 superior
Segmentation mAP@0.5:0.95	0.691	0.769	YOLOv8 superior
High Diagonal Classes	Disease Free, Red Rust	Disease Free	Mask R-CNN more stable
Hard Class	Styler & Root	Styler & Root	Hard for both
Hard Class Recall	0.737	0.900	YOLOv8 better recovery
Hard Class Precision	0.70	0.529	Mask R-CNN more precise
Background Handling	Strong, low FP	Moderate, higher FP	Two-stage advantage
Localization Robustness	Moderate	Strong	YOLOv8 better at strict IoU

4.4 Explainable AI (XAI)

4.4.1 Classification

The ResNet50-DenseNet121 Fusion model had an accuracy of 98.20%, ~0.999 AUC, and MCC of 0.9774. To ensure that the predictions were made on the basis of relevant features of the diseases, four XAI techniques were used on the test set of 111 images: Vanilla Saliency, SmoothGrad, LIME, and SHAP. Vanilla Saliency offered pixel-level explanations but had noisy and edge-centric activations. SmoothGrad removed noise and helped identify regions of lesions. LIME identified superpixels specific to diseases and offered excellent visual explanations. SHAP offered contributions at the feature level and was consistent with high.

ANOVA F-score characteristics, verifying the discriminative ability of embeddings. Overlapping attribution patterns between Phytophthora and Styler & Root were noticed, as expected by errors in the confusion matrix and PCA overlap. SmoothGrad and LIME provided the best spatial attributions, and SHAP verified the importance of features. The experiment verifies that the fusion model is based on lesion patterns rather than background artifacts.

Side-by-side attribution maps for one sample image per class. In Disease Free images, SmoothGrad and SHAP called out uniform leaf texture, while LIME pointed to the entire leaf area as a positive contributor, indicating correct recognition of healthy structural patterns. The vanilla saliency map had more spread-out activity. For the red rust and scab images, the SmoothGrad and LIME saliency maps were able to point out the regions of the lesions, while the SHAP map showed these regions as positive contributions. The vanilla saliency map was able to point out the lesions but with increased noise.In the Phytophthora and Styler & Root images, the activation maps were centered on the infection regions of the lower leaves and the color anomalies. However, the same activation map was discovered for both classes, which accounted for their low misclassification rate.This result is consistent with the PCA analysis and the confusion matrix analysis. When comparing the four XAI techniques on the same image, it can be noticed that gradient-based techniques provide fine-grained but noisy explanations, while chaos-based techniques (LIME and SHAP) provide more stable and interpretable visualizations.

The below analysis of the visual confirms that the fusion model retains its interest in the biologically significant regions of the lesion rather than the background noise, thereby further enhancing the validity of the proposed classification framework.

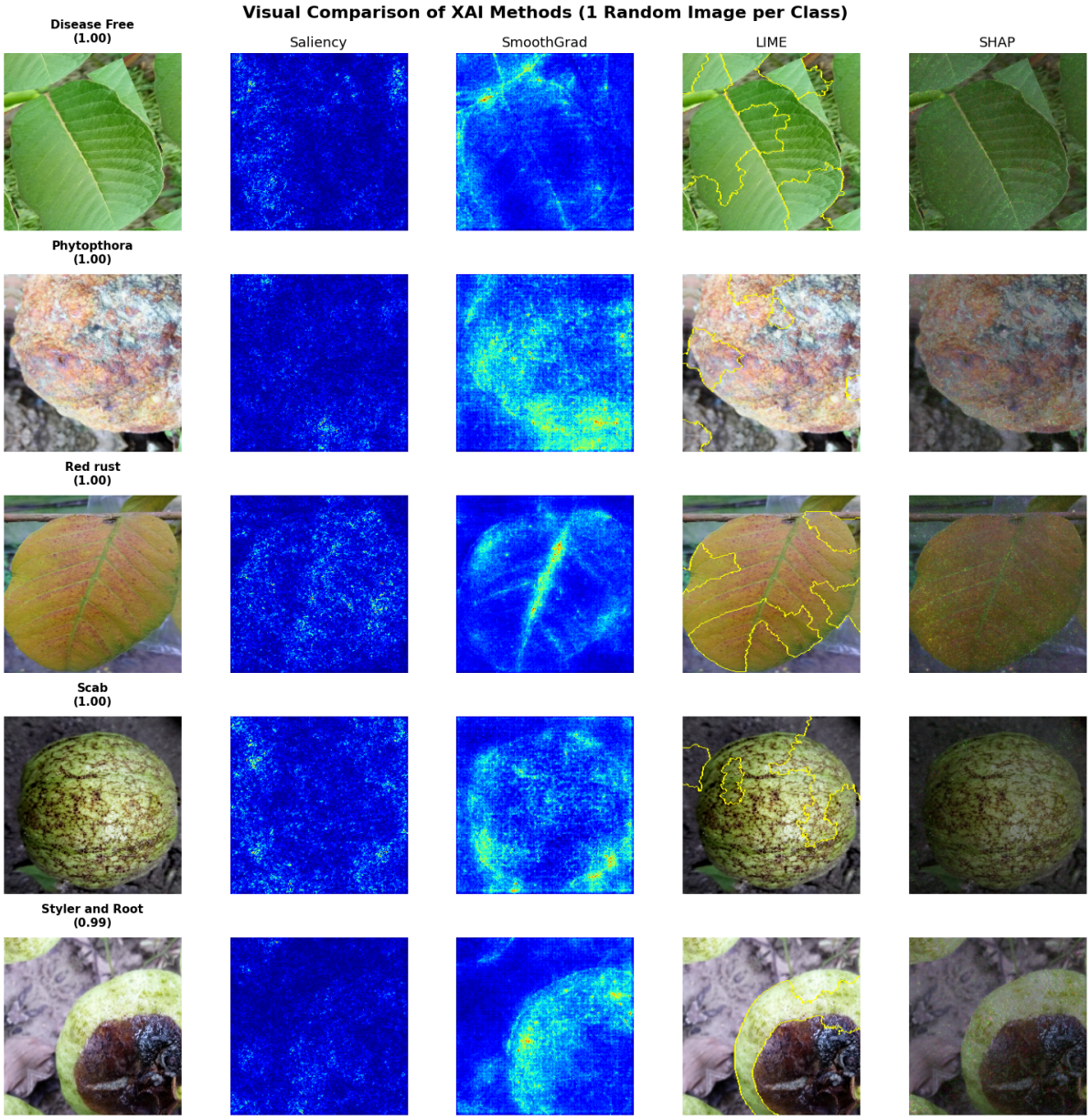
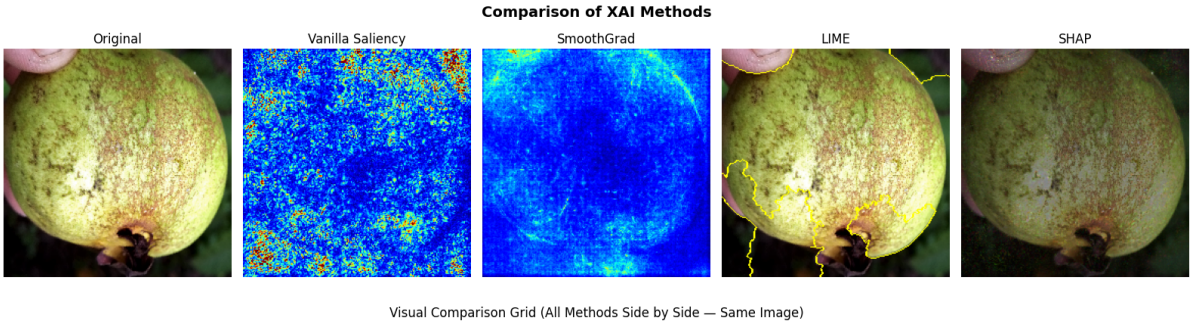


Figure 4.12: Visual Comparison Grid (1 Random Image per Class)



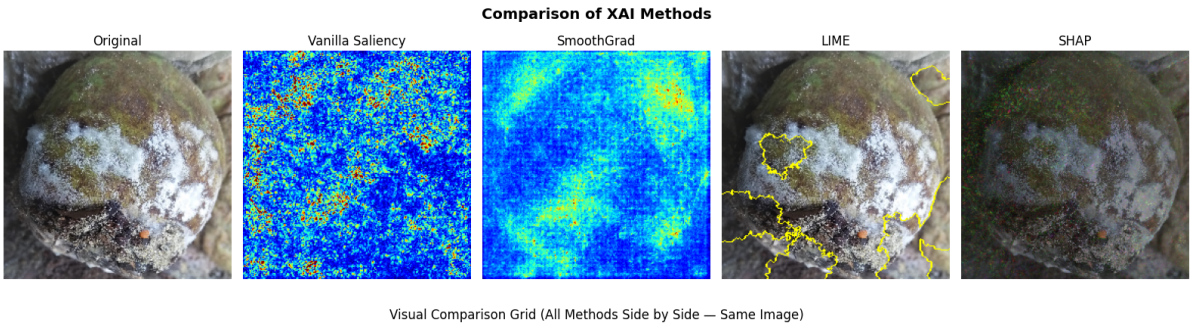


Figure 4.13: Visual Comparison Grid (All Methods Side by Side – Same Image)

4.4.2 Explainable AI (XAI) for detection

Although the deep learning model YOLOv8-seg has high accuracy, it is a black-box model, which makes it difficult to understand the decision-making process. In agricultural disease detection, interpretability is an important aspect that ensures the reliability and trustworthiness of the model. As such, this research combines four XAI techniques: Saliency, SmoothGrad, LIME, and SHAP, to graphically represent the pixel-level contributions that affect the detection and segmentation results. This ensures that the predictions are made based on relevant object areas rather than noise.

4.4.3 Overall Performance

YOLOv8-seg performed better than Mask R-CNN on both detection and segmentation tasks. On detection, YOLOv8-seg had mAP@0.5 of 0.907 and mAP@0.5:0.95 of 0.783, while Mask R-CNN had 0.831 and 0.675, respectively. Although YOLOv8-seg had a similar precision of 0.899 compared to Mask R-CNN (0.900), it had a significantly higher recall of 0.881 compared to 0.587, as well as a higher F1-score of 0.890 compared to 0.711. On segmentation, YOLOv8-seg had mAP@0.5 of 0.889 and mAP@0.5:0.95 of 0.769, which was higher than Mask R-CNN's 0.825 and 0.691, respectively

4.4.4 Visual Comparison of Attribution Maps

Visual comparison (Original → Saliency → SmoothGrad → LIME → SHAP) reveals the presence of significant differences in the quality of explanations. Saliency maps contained noisy activations, SmoothGrad removed noise but did not provide complete object coverage, and LIME provided superpixel-level fragmented explanations. SHAP offered the most spatially well-structured and semantically relevant explanations, which closely matched the bounding box and segmentation maps, emphasizing leaf venation and disease areas.
Fidelity-based ranking: SHAP > SmoothGrad > LIME > Saliency

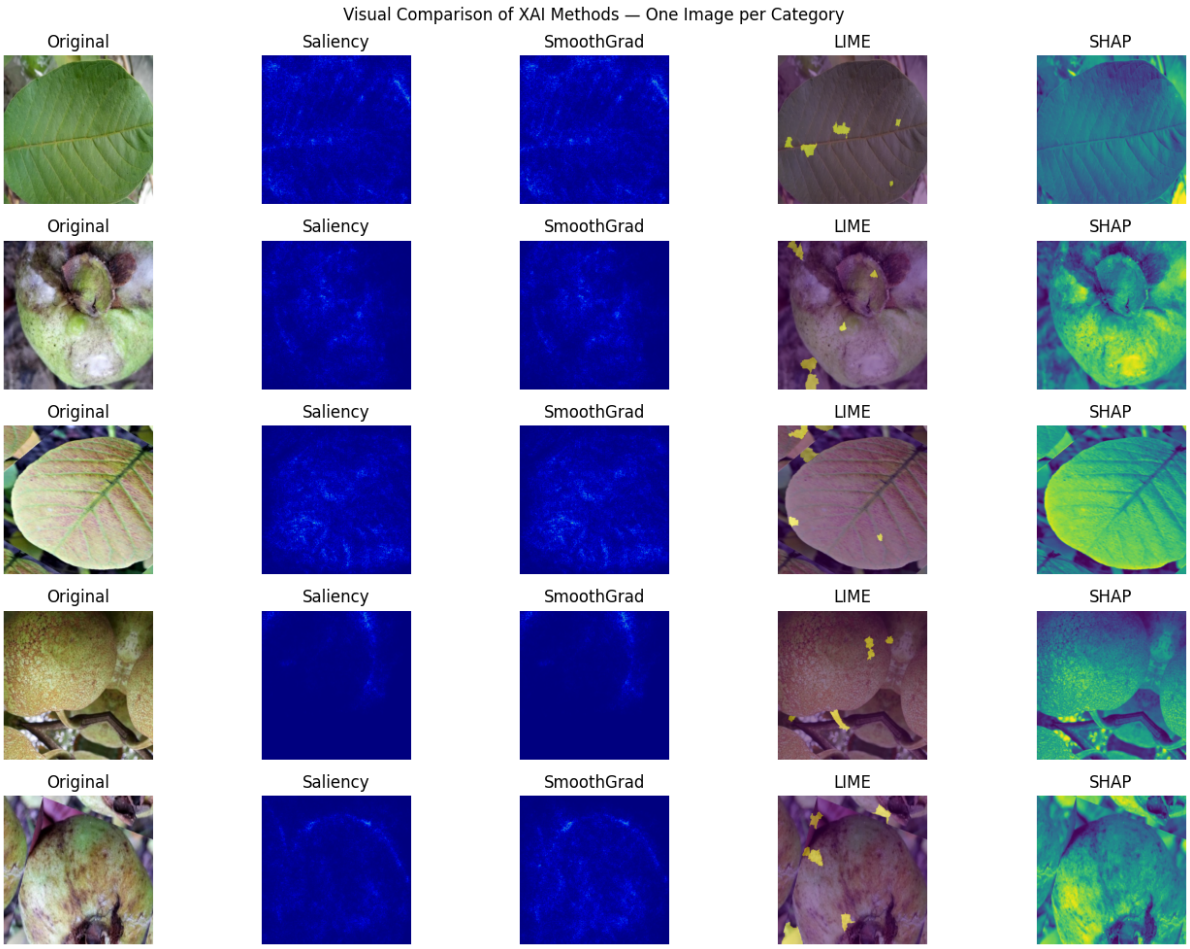


Figure 4.14: Visual Comparison Grid (1 Random Image per Class)

4.5 Deployment Architecture and Web-Based Implementation

This section presents the complete deployment framework of the proposed Guava Fruit and Leaf Disease Analysis System. The system integrates multi-task deep learning models for classification, detection and segmentation within a full-stack web-based client–server architecture. The deployment framework ensures modularity, scalability, computational efficiency, interpretability, and real-time usability for precision agriculture applications. The system supports five guava conditions: Disease Free, Phytophthora, Red Rust, Scab, and Styler & Root. For classification, the best- performing architecture is the ResNet50–DenseNet121_ Fusion model, which provides high discriminative capability by combining global semantic features and fine-grained texture information. For detection and segmentation, the system utilizes YOLOv8-seg, which enables simultaneous bounding box localization and pixel-level mask generation within a unified framework. Each uploaded image undergoes structured multi-stage inference, generating separate outputs for global prediction, spatial localization, pixel-level delineation, and model interpretability.

4.5.1 System Architecture

The proposed system is based on a full-stack client-server architecture, in which both the frontend and backend parts work together to accomplish the task of disease analysis and structured visualization. Unlike the standalone experimental models, the implemented system combines user interaction, real-time communication, inference computation, and visualization in a single pipeline. The architecture consists of the following layers:

- Client Layer (Web Browser Interface)
- Frontend Application Layer
- REST API Communication Layer
- Backend Inference Engine
- Deep Learning Model Layer
- Visualization Layer

The layers are designed in such a way that there is a separation of concerns. This implies that modifications can be done in the model independently of the user interface.

4.5.2 Frontend Architecture

The frontend component is also extremely important, apart from the visualization part. It offers a friendly graphical user interface (GUI) for image upload and interaction. Before the image is transmitted, format and size validation is done by validation tools on the client-side to avoid unnecessary computation on the server side. The frontend communicates with the backend using asynchronous REST API calls (for example, Fetch/AJAX) to enable non-blocking execution. After the frontend has access to the structured JSON output, it dynamically displays the predicted label for the disease with confidence, overlays the detection bounding boxes and displays the segmentation masks. The structured display enables transparent decision-making.

4.5.3 Backend Architecture

The backend is the computational inference engine that is responsible for the execution of the multi-task deep learning pipeline. After the image has been uploaded through a REST endpoint, standardized preprocessing tasks such as resizing, normalization, and tensor conversion are performed. The processed input is then passed through the classification, detection, segmentation, and explain ability modules in a sequential manner. The classification neural network predicts the disease class along with the corresponding probability scores. The detection model performs spatial localization through bounding box generation. The segmentation model offers pixel-level demarcation of the infected areas. All outputs are combined into a structured JSON response and sent to the frontend for visualization. GPU acceleration is optionally used to support efficient real-time inference.

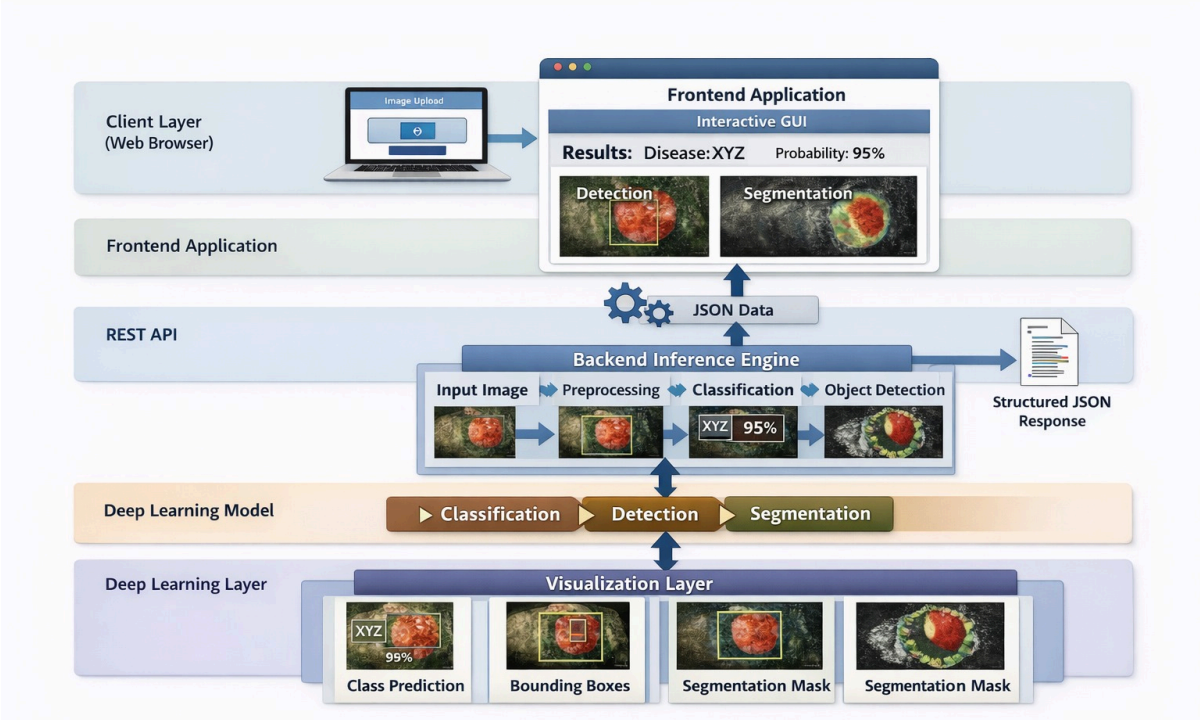


Figure 4.15: Automated disease analysis

4.5.4 Backend Implementation

The backend is written using a Python-based REST framework like FastAPI or Flask. The backend is responsible for handling HTTP requests and performing deep learning inference.

Technologies Used

- Python (Core Programming Language)
- TensorFlow / PyTorch (Deep Learning Framework)
- OpenCV (Image Processing)
- NumPy (Numerical Computation)
- FastAPI / Flask (REST API Framework)
- Optional GPU Acceleration for inference optimization

The backend accepts the uploaded image via a POST endpoint. The image is then resized to conform to model input sizes, pixel values normalized, and the image transformed into tensor representation. The fusion model integrates feature spaces from ResNet50 and DenseNet121 Backbones via Global Average Pooling and feature concatenation. The fully connected layer with dropout and soft max activation function provides the predicted class and confidence level of the disease. There is an object detection model (yolov8-seg) that detects the infected region using bounding boxes and provides the coordinates, class, and confidence level. The semantic segmentation model provides the infected region probability map. A Grad-CAM or activation-based method provides attention heat maps that focus on regions of interest for classification. The output is organized in JSON format with class prediction, confidence scores, detection and segmentation masks.

4.5.5 Frontend Implementation

The frontend is built using conventional web technologies to make it accessible and dynamically interactive.

Technologies Used

- HTML (Structural Layout)
- CSS (Styling and Design)
- JavaScript (Dynamic Interaction)
- Fetch/AJAX API Calls (Backend Communication)

The frontend has the capability of file selection and drag-and-drop functionality. It ensures proper file format and size compliance. It transmits the image to the backend and receives inference outputs without reloading the page. It presents the classification label with confidence, detection bounding boxes and segmentation masks in distinct structured panels. The structured visualization makes it more interpretable and user-friendly.

4.6 User Interface Design

The user interface is designed to ensure clarity, transparency, and usability for agricultural practitioners and researchers. The UI components include the header section displaying the system title , image upload panel , original image preview window , four separate result panels: classification Output , detection output and segmentation output confidence score display. Once the image is uploaded, the preview is displayed, and a processing indicator is also displayed. After the backend inference is completed, the results are dynamically displayed. The bounding boxes and segmentation results are displayed along with the classification result.

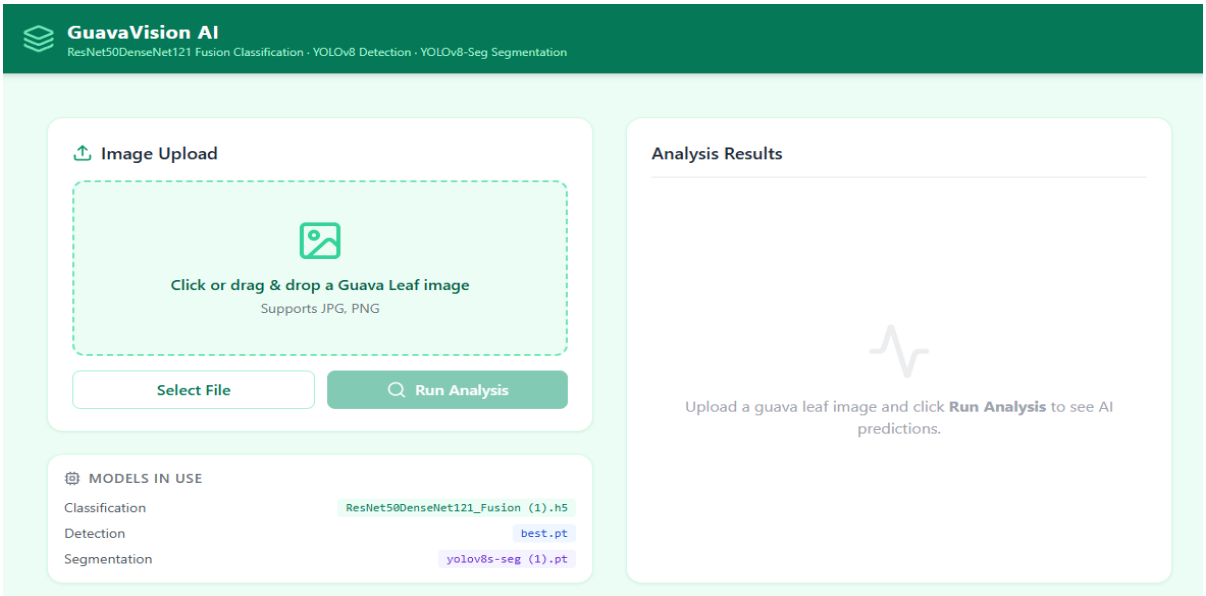


Figure 4.16: GuavaGuard AI web application user interface (before prediction)

4.7 Classification Analysis (Five Classes)

For Disease Free Healthy leaves and fruits are correctly identified with high confidence. Absence of disease-specific patterns leads to reliable healthy classification. In the case of Phytophthora, distinct infection patterns enable accurate differentiation from other disease classes. Similarly Red Rust is effectively recognized through texture irregularities and color variations characteristic of rust infections are successfully captured by learned feature representations. Moreover, distinguished based Scab infection are on lesion-specific structural patterns contribute to reliable discrimination from other diseases. Likewise, Styler & Root disease is identified through Structural deformities and infection-specific characteristics enable correct classification. Confidence scores associated with predictions confirm probabilistic reliability.

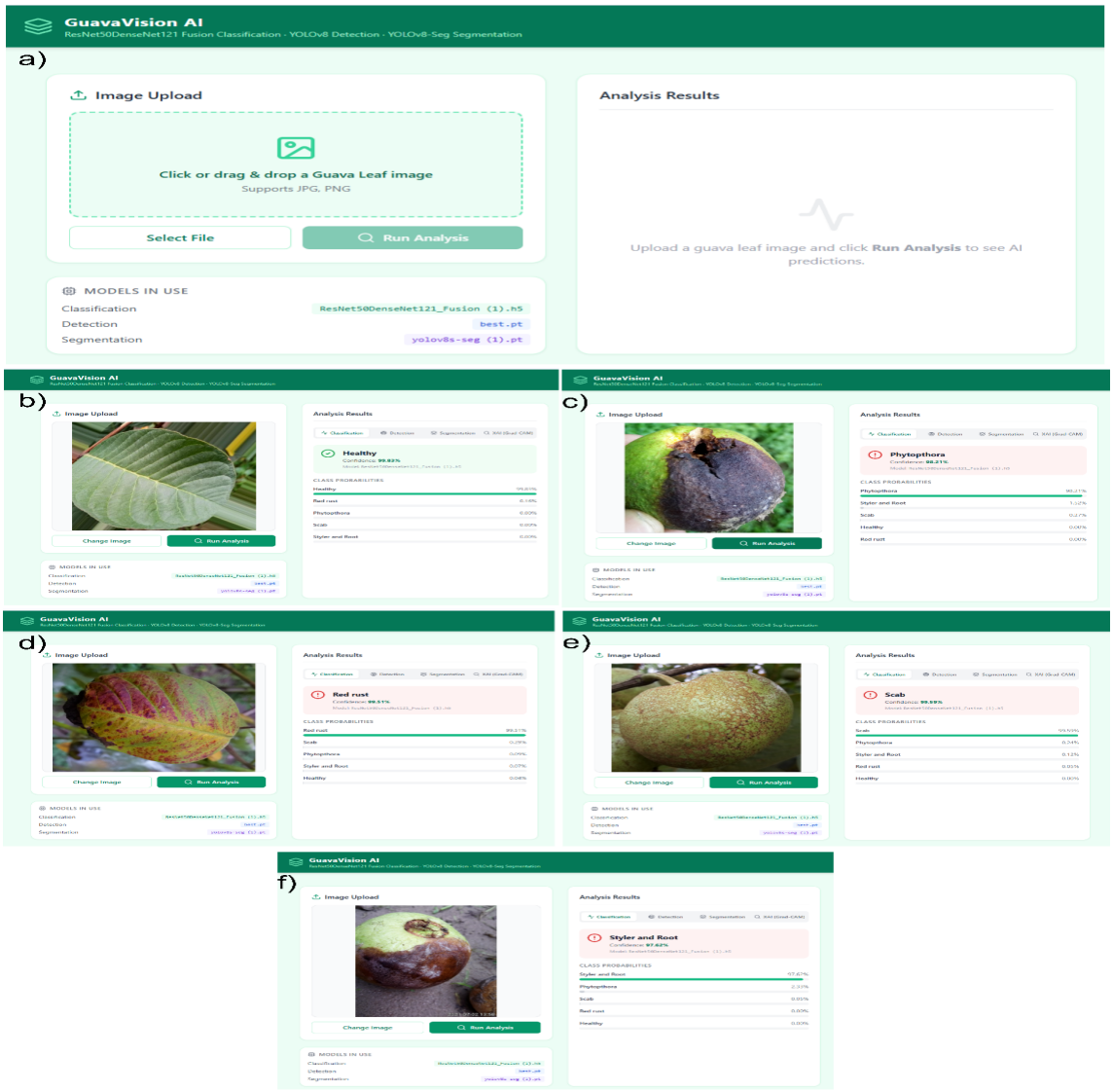


Figure 4.17: Web Based Classification Output for Five Guava Disease Classes

4.8 Detection Analysis (Five Classes)

The detection module performs spatial localization of infected regions using bounding boxes. For Disease Free Minimal or no bounding boxes are generated, indicating low false-positive rates. In the case of Phytophthora, infected regions are accurately localized with high confidence. Similarly in Red Rust multiple infection spots are detected and enclosed within separate bounding boxes. Moreover, distinguished based Scab infection are on lesion areas are clearly marked with precise bounding box placement. Likewise,in Styler & Root Structural infection regions are effectively localized. Detection confidence scores reflect model certainty in region prediction.

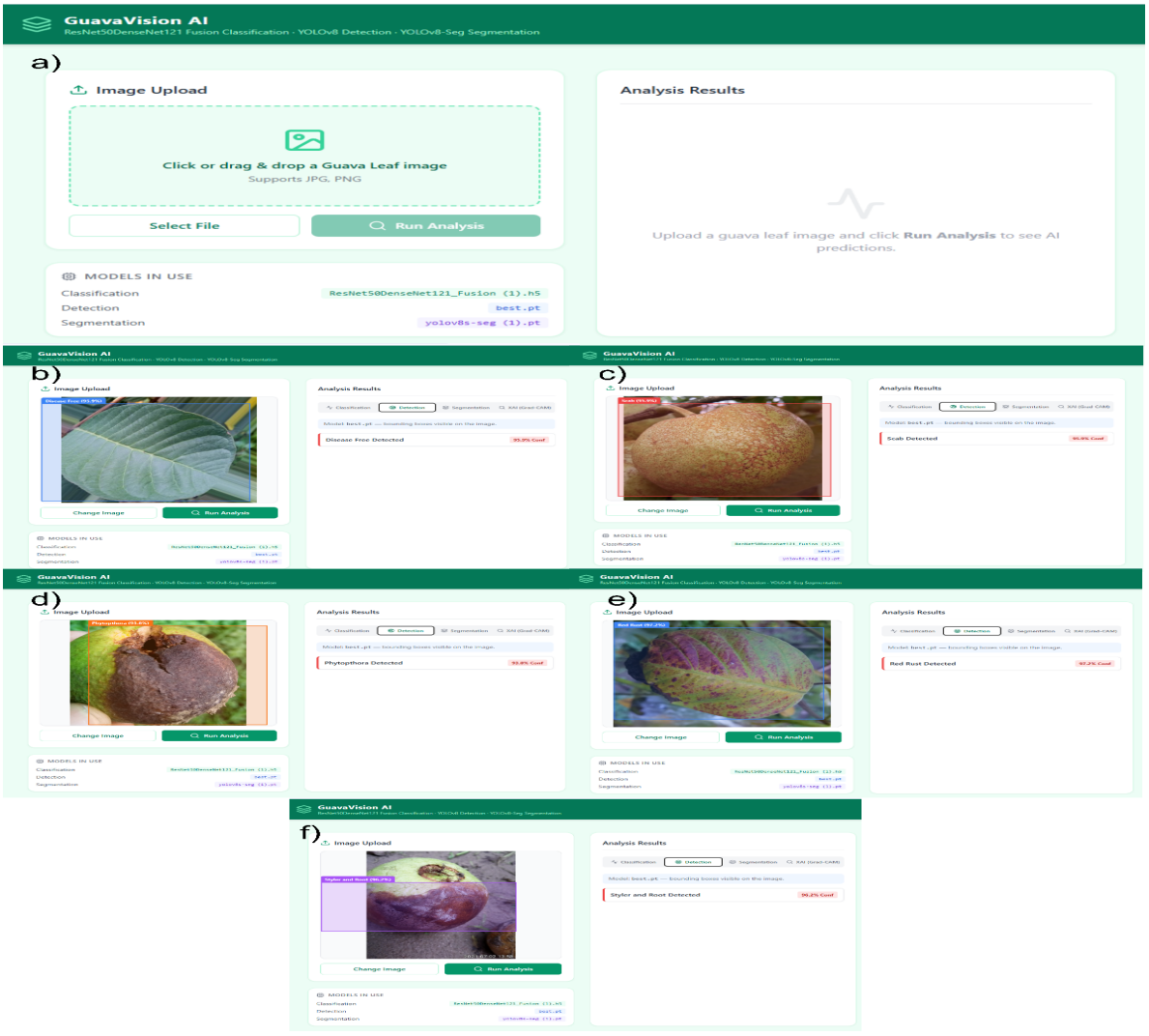


Figure 4.18: Web Based Detection Output for Five Guava Disease Classes

4.9 Segmentation Analysis (Five Classes)

The segmentation module provides pixel-level disease delineation. Disease Free: Segmentation masks are minimal or absent, confirming healthy samples.

Phytophthora: regions are precisely segmented at the pixel level.
Red Rust: Rust patches are distinctly separated from healthy tissue.
Scab: Clear boundary separation between diseased and healthy regions is observed.
Styler & Root: Structural infection areas are accurately highlighted.
Segmentation enhances disease severity assessment beyond rectangular detection.

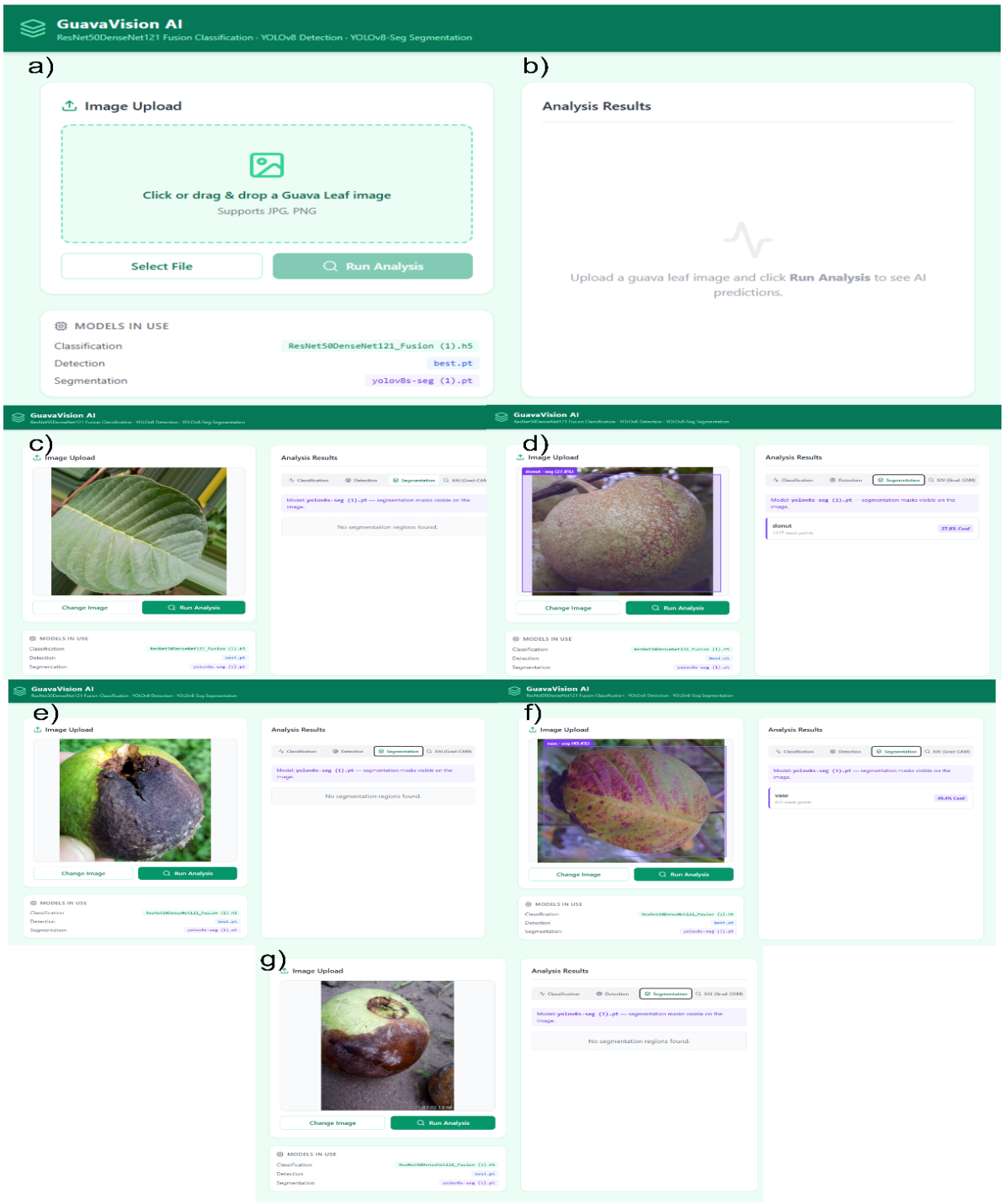


Figure 4.19: Web Based Segmentation Output for Five Guava Disease Classes

The deployed system showcases a full-stack multi-task deep learning framework that encompasses classification, detection, segmentation, and explainability in a web-based scalable architecture. The frontend is responsible for user interaction, validation, asynchronous communication, and structured visualization, while the backend handles computational inference and response generation. This helps in achieving efficiency, maintainability, and scalability. The combined architecture offers global disease recognition, localization, fine-grained segmentation, and transparent reasoning in a single processing system. The system is applicable for real-time guava disease analysis and precision agriculture.

Chapter 5 Conclusion

5.1 Conclusion

This research work developed an end-to-end deep learning model for the classification, detection, and segmentation of guava fruit and leaf diseases using explainable AI and web-based implementation. The ResNet50 + DenseNet121 Fusion model achieved the highest accuracy of 98.20% in classification, establishing the success of feature fusion. The performance was adequate for individual performance on DenseNet121, and MobileNetV2 was appropriate for lightweight web deployment. The output of Grad-CAM validated the focus of the classification network on biologically significant regions of lesions, and interpretability and validity were ensured. Detection and segmentation were enhanced by YOLOv8-seg with enhanced mAP and faster inference times, and Mask R-CNN with increased precision and decreased false positives. Analysis of SHAP explanation showed that the correct localization of lesions in the segmentation outcome is valid. Although the results are excellent, some drawbacks are the small number of samples in the dataset, the similarity of some classes of diseases, and the lack of cross-validation. Future studies will focus on larger datasets in various settings, transformer hybrid models, disease severity assessment, and mobile applications.

5.2 Future Work

Future research will concentrate on the following: extending the dataset using field data, enhancing the segmentation algorithm using sophisticated architectures, developing lightweight models for edge devices, and incorporating multi-modal data for early disease diagnosis. Moreover, future systems can also incorporate disease severity prediction and cross-crop generalization to improve agricultural applications.

5.3 Challenges

Some of the challenges faced during the course of this research include the small size of the dataset, imbalance in the classes, and the difficulties in creating accurate annotations for segmentation. The original dataset had only 527 images, which made it difficult for the models to generalize and even had the potential for overfitting. There was also confusion in the classification of certain classes of diseases due to their visual similarity.

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